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Ocenjevanje volatilnosti z Markovskim prehajanjem na finančnih trgih s sekvenčnimi Monte Carlo metodami

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Worth mentioning in the acknowledgment is everyone who contributed to your thesis.

Vito Levstik, 2026

To all the flowers of this world.

"The only reason for time is so that everything doesn't happen at once."

— Albert Einstein

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List of used acronyms

acronym	meaning
CA	classification accuracy
DBMS	database management system
SVM	support vector machine
...	...

Abstract

Title: Estimating Markov switching volatility in financial markets using sequential Monte Carlo methods

This sample document presents an approach to typesetting your BSc thesis using LaTeX. A proper abstract should contain around 100 words which makes this one way too short. A good abstract contains: (1) a short description of the tackled problem, (2) a short description of your approach to solving the problem, and (3) (the most successful) result or contribution in your thesis.

Keywords

computer, computer, computer

Povzetek

Naslov: Ocenjevanje volatilnosti z Markovskim prehajanjem na finančnih trgih s sekvenčnimi Monte Carlo metodami

V vzorcu je predstavljen postopek priprave magistrskega dela z uporabo okolja LaTeX. Vaš povzetek mora sicer vsebovati približno 100 besed, ta tukaj je odločno prekratek. Dober povzetek vključuje: (1) kratek opis obravnavanega problema, (2) kratek opis vašega pristopa za reševanje tega problema in (3) (najbolj uspešen) rezultat ali prispevek magistrske naloge.

Ključne besede

računalnik, računalnik, računalnik

Chapter 1

Introduction

Chapter 2

Stochastic Volatility

Volatility is a fundamental concept in financial econometrics and is commonly used as a measure of the variability or uncertainty of asset returns. In its most basic form, volatility refers to the standard deviation of returns over a given time horizon, and therefore quantifies the magnitude of fluctuations around the expected return.

Many models assume that volatility is constant. In practice, however, financial markets exhibit periods of high volatility, for example, during financial crises, and periods of low volatility, usually during stable economic conditions. Stochastic volatility models capture this behavior by allowing volatility to vary over time according to a stochastic process.

This chapter begins by introducing volatility and then develops the state-space formulation used throughout the thesis, with particular emphasis on the Markov switching stochastic volatility (MSSV) model.

2.1 Volatility

Let P_t denote the price of an asset (stock, index, commodity, etc.) at time t.

Definition 2.1.1. The *log return* of the asset between two consecutive time

points is defined as

$$Y_t = \log(P_t) - \log(P_{t-1}) = \log\left(\frac{P_t}{P_{t-1}}\right). \tag{2.1.1}$$

The advantage of using log returns rather than simple returns lies in their additive structure over time and their natural treatment of compounding effects [1]. Throughout this thesis, the term return will also refer to a log return.

Figure 2.1.1 shows the log returns of the S&P 500 index from 1980 till mid 2026. Clearly the data exhibits heteroscedasticity, as the variance is not constant over time. In fact, we can see clusters of high and low variability, which will motivate the use of multiple regimes to model the volatility process.

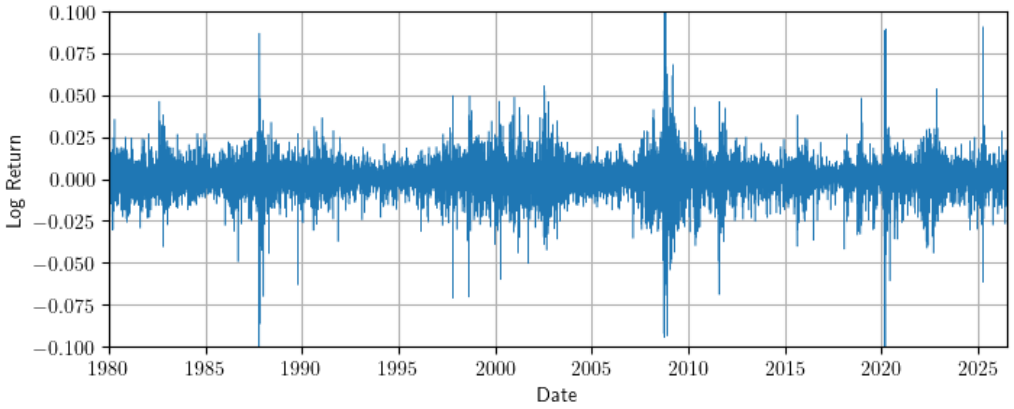


Figure 2.1.1: Log returns of the S&P 500 index from 1980 till mid 2026.

While returns are directly observable, volatility is a characteristic of the underlying return distribution and is not directly observable from a single realization. At each time point t, only one return is observed, whereas volatility is defined through the variability of the entire return-generating distribution. Consequently, time-varying volatility must be inferred from data rather than measured directly.

Definition 2.1.2. Let $\mathcal{F}_{t-1}$ denote the information set containing all relevant information available up to time $t-1$. The *conditional mean* and *conditional*

variance of the return are defined as

$$\mu_t = E[Y_t \mid \mathcal{F}_{t-1}], \tag{2.1.2}$$

$$\sigma_t^2 = \mathrm{Var}(Y_t \mid \mathcal{F}_{t-1}) = E[(Y_t - \mu_t)^2 \mid \mathcal{F}_{t-1}], \tag{2.1.3}$$

and σ_t is referred to as the *conditional volatility* [2].

In many financial applications, the conditional mean is assumed to be negligible relative to the variability of returns, so that the focus is primarily on modelling the conditional volatility process.

So far, we have not specified the time scale of the returns. For example, Y_t could represent daily, weekly, or monthly returns, and t could therefore index days, weeks, or months. In this thesis, we focus on daily returns, specifically close-to-close daily returns, meaning that P_t is the closing price of the asset on day t[1]. The goal is then to estimate the daily conditional volatilities σ_t.

2.1.1 Realized variance

When high-frequency intraday data are available, the latent conditional variance can be approximated using realized variance. Consider a day t, divided into m equally spaced intraday intervals. Let $Y_{t,i}^{(m)}$ denote the return over the i-th intraday interval, $i = 1, \ldots, m$, so that the corresponding daily return can be written as

$$Y_t = \sum_{i=1}^{m} Y_{t,i}^{(m)}. \tag{2.1.4}$$

Definition 2.1.3. The *realized variance* of day t at sampling frequency m is defined as

$$RV_t^{(m)} = \sum_{i=1}^{m} \left(Y_{t,i}^{(m)}\right)^2. \tag{2.1.5}$$

It is often reasonable to assume that

$$E\left[Y_{t,i}^{(m)} \mid \mathcal{F}_{t-1}\right] \approx 0, \tag{2.1.6}$$

$$\mathrm{Cov}\left(Y_{t,i}^{(m)}, Y_{t,j}^{(m)} \mid \mathcal{F}_{t-1}\right) \approx 0, \qquad \text{for } i \neq j. \tag{2.1.7}$$

[1] One could also consider open-to-open returns, where P_t is the opening price of the asset on day t.

Under these assumptions, we have

$$\sigma_t^2 = \mathrm{Var}(Y_t \mid \mathcal{F}_{t-1}) = \mathrm{Var}\left(\sum_{i=1}^{m} Y_{t,i}^{(m)} \,\middle|\, \mathcal{F}_{t-1}\right) \tag{2.1.8}$$

$$\approx \sum_{i=1}^{m} \mathrm{Var}\left(Y_{t,i}^{(m)} \mid \mathcal{F}_{t-1}\right) \tag{2.1.9}$$

$$\approx \sum_{i=1}^{m} E\left[\left(Y_{t,i}^{(m)}\right)^2 \,\middle|\, \mathcal{F}_{t-1}\right] \tag{2.1.10}$$

$$= E\left[RV_t^{(m)} \mid \mathcal{F}_{t-1}\right]. \tag{2.1.11}$$

Thus, under these assumptions, realized variance provides an approximately unbiased estimator of the conditional variance. Moreover, as the sampling frequency increases, the realized variance generally becomes a more precise measure of the underlying volatility, since the estimation error decreases with m. In practice, however, m must not be too large, to prevent intraday returns from becoming correlated due to market microstructure effects [3].

2.1.2 Overnight returns

Many assets are not traded continuously, as markets are closed overnight, over weekends, and on holidays. We therefore decompose the daily return into overnight (close-to-open) and intraday (open-to-close) components,

$$Y_t = \log\left(\frac{P_t^{close}}{P_{t-1}^{close}}\right) = \log\left(\frac{P_t^{open}}{P_{t-1}^{close}}\right) + \log\left(\frac{P_t^{close}}{P_t^{open}}\right) = Y_t^{CO} + Y_t^{OC} \tag{2.1.12}$$

where Y_t^{CO} is the overnight (close-to-open) return and Y_t^{OC} is the intraday (open-to-close) return. Here we see the usefulness of the additive structure of log returns. From this it naturally follows that the daily variance can be decomposed into an overnight and an intraday component, if we assume that the two components are uncorrelated:

$$\sigma_t^2 = \mathrm{Var}(Y_t \mid \mathcal{F}_{t-1}) = \mathrm{Var}(Y_t^{CO} + Y_t^{OC} \mid \mathcal{F}_{t-1}) \tag{2.1.13}$$

$$= \mathrm{Var}(Y_t^{CO} \mid \mathcal{F}_{t-1}) + \mathrm{Var}(Y_t^{OC} \mid \mathcal{F}_{t-1}). \tag{2.1.14}$$

While the intraday variance can be estimated from high-frequency data using realized variance $RV_t^{(m)}$, the overnight variance is difficult to estimate reliably from a single observation[2]. One solution is to simply add the squared overnight return to $RV_t^{(m)}$. The variance estimator is then given by

$$\hat{\sigma}_t^2 = RV_t^{(m)} + (Y_t^{CO})^2. \tag{2.1.15}$$

This approach is problematic, as the squared overnight return is a very noisy estimator of the overnight variance [4].

A better approach is to scale the intraday realized variance, which gives the estimator

$$\hat{\sigma}_t^2 = cRV_t^{(m)}, \qquad c = \frac{\frac{1}{T}\sum_{t=1}^{T}(Y_t - \hat{\mu}_t)^2}{\frac{1}{T}\sum_{t=1}^{T} RV_t^{(m)}}, \tag{2.1.16}$$

where $\hat{\mu}_t$ is the sample mean of the daily returns and T is the total number of days in the sample. Here, c measures the historical ratio of total close-to-close variance to intraday variance. Equivalently, if κ denotes the historical ratio of overnight to intraday variance, then $c = 1 + \kappa$. Thus, the missing overnight variance is estimated as a proportion κ of the intraday realized variance, rather than from the noisy overnight return of each individual day. Such an approach is valid under the assumption that intraday and overnight returns are uncorrelated. Weekends and holidays are naturally treated as extended overnight periods [3, 4].

Overall, when high-frequency intraday observations are available, they provide a direct and increasingly precise proxy for the underlying daily variance. In empirical applications, realized variance can therefore be used in place of the unobserved conditional variance, which provides a way to evaluate the performance of volatility models.

[2]In case when high-frequency overnight trading data is available, the overnight variance can be approximated in the same way using intranight squared returns, just like the intraday variance [4].

2.2 State-space models

A state-space model describes a dynamical system in which an unobserved (latent) stochastic process drives the evolution of an observed process. The central idea is that the system evolves through hidden states, while only noisy measurements of those states are available.

Let $\{X_t\}_{t\geq 0}$ denote the sequence of random variables representing the latent state process, let $\{Y_t\}_{t\geq 1}$ denote the sequence of random variables representing the observed process, and let T denote the time horizon, hence $t \in \{0, 1, \ldots, T\}$[3]. X_t takes values in a measurable state space $(\mathcal{X}, \mathcal{B}(\mathcal{X}))$ and Y_t takes values in $(\mathcal{Y}, \mathcal{B}(\mathcal{Y}))$. Together, $\{(X_t, Y_t)\}_{t\geq 1}$ forms a stochastic process on the joint space $(\mathcal{X} \times \mathcal{Y}, \mathcal{B}(\mathcal{X}) \otimes \mathcal{B}(\mathcal{Y}))$ [5].

Definition 2.2.1. A *state-space model* with parameters $\theta \in \Theta$ is a pair of processes $\{X_t\}_{t\geq 0}$ and $\{Y_t\}_{t\geq 1}$ such that

1. the latent process $\{X_t\}_{t\geq 0}$ is a Markov process,
2. the observations $\{Y_t\}_{t\geq 1}$ are conditionally independent given the latent process, and Y_t depends on the latent process only through X_t,

specified by the densities

$$X_0 \sim \chi(x_0 \mid \theta), \tag{2.2.1}$$

$$X_t \mid X_{t-1} = x_{t-1} \sim p(x_t \mid x_{t-1}, \theta), \tag{2.2.2}$$

$$Y_t \mid X_t = x_t \sim f(y_t \mid x_t, \theta), \tag{2.2.3}$$

where χ is the *initial density*, p is the *transition density*, and f is the *observation* or *emission density* [5, 6, 7].

Remark 2.2.1. The conditional distribution of $X_t \mid X_{t-1}$ need not admit a density in general. A proper way would be to define the state-space model

[3]The latent states are indexed from 0 to T, whereas the observed returns are indexed from 1 to T. This emphasizes that the initial state X_0 is the prior and is not associated with any observation. The indexing could be defined differently (e.g., with observations also starting at $t = 0$), but we adopt this convention throughout.

in a more measure focused way using probability kernels [5]. However, we will not deal with this more general definition in this thesis, as our models admit densities.

Sometimes, the state-space model is defined in even more general terms, where the latent process and observation process are also dependent on the control input U_t, which constitutes another sequence of random variables [8]. Such a definition is more common in control theory and engineering. The control inputs can be thought of as external factors that influence the system dynamics and observations. In this thesis, we will not consider control inputs, as we are primarily interested in modelling the latent volatility process and its relationship with observed returns.

The main goal of state-space models is to estimate the latent states based on the observed data. With $X_{i:j} = \{X_i, X_{i+1}, \dots, X_j\}$ we denote the sequence of random variables from time i to j, and with $x_{i:j} = \{x_i, x_{i+1}, \dots, x_j\}$ we denote their realizations.

Definition 2.2.2. Let $p(x_{0:t}, \theta \mid y_{1:t})$ be the joint posterior distribution of the latent states and model parameters given the observed data. We define

- the *smoothing distribution*

$$p(x_{0:t} \mid y_{1:t}) = \int p(x_{0:t}, \theta \mid y_{1:t}) \, \mathrm{d}\theta, \tag{2.2.4}$$

- the *filtering distribution*

$$p(x_t \mid y_{1:t}) = \int p(x_{0:t}, \theta \mid y_{1:t}) \, \mathrm{d}x_{0:t-1} \, \mathrm{d}\theta, \tag{2.2.5}$$

- the *predictive distribution*

$$p(x_{t+1} \mid y_{1:t}) = \int p(x_{t+1} \mid x_t, \theta) p(x_{0:t}, \theta \mid y_{1:t}) \, \mathrm{d}x_{0:t} \, \mathrm{d}\theta, \tag{2.2.6}$$

- the *posterior distribution of the parameters*

$$p(\theta \mid y_{1:t}) = \int p(x_{0:t}, \theta \mid y_{1:t}) \, \mathrm{d}x_{0:t}. \tag{2.2.7}$$

When the parameters θ are known, the smoothing, filtering and predictive distributions are defined analogously, by conditioning on θ instead of integrating it out, i.e., $p(x_{0:t} \mid y_{1:t}, \theta)$, $p(x_t \mid y_{1:t}, \theta)$ and $p(x_{t+1} \mid y_{1:t}, \theta)$.

Definition 2.2.3. A *hidden Markov model* (HMM) is a state-space model whose latent state takes values in a finite set $\mathcal{X} = \{1, 2, \ldots, K\}$, where K is the number of states.

Although there is some discrepancy in the literature, and the term hidden Markov model is sometimes used for state-space models in general, we adopt the definition above [5].

2.3 Markov switching stochastic volatility model

The Markov switching stochastic volatility (MSSV) model is the main model of interest in this thesis. It is a state-space model in which the hidden state is the tuple $X_t = (H_t, S_t)$. The log volatility H_t is continuous, while the regime indicator S_t is discrete and takes values in $\{1, 2, \ldots, K\}$, where K is the number of regimes [9].

Definition 2.3.1. The *Markov switching stochastic volatility model* with K regimes is the state-space model with latent state $X_t = (H_t, S_t)$ taking values in

$$\mathcal{X} = \mathbb{R} \times \{1, 2, \ldots, K\}, \tag{2.3.1}$$

equipped with the product σ-algebra $\mathcal{B}(\mathbb{R}) \otimes 2^{\{1,2,\ldots,K\}}$, and the following distributions [9, 10].

- **Observation distribution:**

$$Y_t \mid X_t = (h_t, s_t) \sim \mathcal{N}\left(0, \exp\left(h_t\right)\right). \tag{2.3.2}$$

- **Transition distribution:**

$$H_t \mid H_{t-1} = h_{t-1}, S_t = s_t \sim \mathcal{N}(\mu_{s_t} + \phi(h_{t-1} - \mu_{s_t}), \eta^2), \tag{2.3.3}$$

$$S_t \mid S_{t-1} = s_{t-1} \sim \text{Categorical}([p_{s_{t-1},1}, \ldots, p_{s_{t-1},K}]), \tag{2.3.4}$$

  where the regime S_t is conditionally independent of $H_{0:t-1}$ and $S_{0:t-2}$ given S_{t-1}, and the log volatility H_t is conditionally independent of $H_{0:t-2}$ and $S_{0:t-1}$ given H_{t-1} and S_t.

- **Initial distribution:**

$$H_0 \mid S_0 = s_0 \sim \mathcal{N}\left(\mu_{s_0}, \frac{\eta^2}{1-\phi^2}\right), \tag{2.3.5}$$

$$S_0 \sim \text{Categorical}\left(\frac{1}{K}, \frac{1}{K}, \ldots, \frac{1}{K}\right). \tag{2.3.6}$$

Here $\mathcal{N}$ denotes the normal distribution with the given mean and variance, μ_s is the regime-specific mean of regime s, ϕ is the persistence parameter, η^2 is the variance of the log-volatility innovation process, and $p_{ij} = \mathbb{P}(S_t = j \mid S_{t-1} = i)$ is the probability of transitioning from regime i to regime j. The transition probabilities are collected in the $K \times K$ row-stochastic regime transition matrix

$$P = \begin{bmatrix} p_{11} & p_{12} & \cdots & p_{1K} \\ p_{21} & p_{22} & \cdots & p_{2K} \\ \vdots & \vdots & \ddots & \vdots \\ p_{K1} & p_{K2} & \cdots & p_{KK} \end{bmatrix}, \qquad \sum_{j=1}^{K} p_{ij} = 1 \quad \text{for all } i. \tag{2.3.7}$$

The observation distribution admits a density function, which we denote by $f(y_t \mid x_t, \theta) = f(y_t \mid h_t)$, as it does not depend on the regime s_t or on any of the model parameters θ. Note that mathematically speaking the latent state H_t is the log variance of the return Y_t, but it is often referred to as the log volatility in the literature [10, 11, 12]. We will also follow this convention and refer to H_t as the log volatility.

Observe that, given the regimes, the log volatility process is essentially an AR(1) process with regime-specific means. The initial distribution of H_0 is

chosen as the stationary distribution of the corresponding AR(1) component, and the initial regime has a uniform prior [9]. The initial distribution admits a density function, which can be written as $\chi(x_0 \mid \theta) = \chi(s_0 \mid \theta)\chi(h_0 \mid s_0, \theta)$.

At first, it may seem that such a conditional definition breaks the Markov property of the latent state process, as H_t depends on S_t. The following proposition shows that this is not the case.

Proposition 2.3.1. *The latent process* $\{X_t\}_{t\geq 0} = \{(H_t, S_t)\}_{t\geq 0}$ *of the MSSV model is a Markov process with transition density*

$$p(x_t \mid x_{t-1}, \theta) = p(s_t \mid s_{t-1}, \theta)\, p(h_t \mid h_{t-1}, s_t, \theta). \tag{2.3.8}$$

Proof. Using the definition of conditional probability

$$\begin{aligned} p(x_t \mid x_{t-1}, \theta) &= p((h_t, s_t) \mid (h_{t-1}, s_{t-1}), \theta) && (2.3.9)\\ &= p(h_t \mid h_{t-1}, s_{t-1}, s_t, \theta)\, p(s_t \mid h_{t-1}, s_{t-1}, \theta) && (2.3.10)\\ &= p(h_t \mid h_{t-1}, s_t, \theta)\, p(s_t \mid s_{t-1}, \theta). && (2.3.11) \end{aligned}$$

□

In order for the MSSV model to be well-defined, the following assumptions are made.

1. The persistence ϕ lies in the range $(-1, 1)$.
2. Each regime corresponds to at least one time point.
3. The regime means are ordered such that $\mu_1 < \mu_2 < \ldots < \mu_K$.

Assumption 1 ensures that the AR(1) process is stationary, as shown in the next proposition. This is needed to ensure that the log volatility does not diverge to infinity. Assumption 2 is necessary to avoid empty regimes, as otherwise some μ_i would be unidentified. Assumption 3 is a common identification constraint in mixture models, which prevents label switching and ensures that the regimes are uniquely defined [9].

In this thesis, we ensure the identifiability of the regimes using the following parameterization:

$$\mu_i = \mu_{i-1} + e^{\delta_i}, \qquad i = 2, \ldots, K, \tag{2.3.12}$$

where $\delta_i \in \mathbb{R}$. This parameterization is motivated by the fact that the exponential function is strictly positive, which guarantees that $\mu_i > \mu_{i-1}$ while leaving δ_i unconstrained.

The parameters of the MSSV model are then given by the vector

$$\theta = \{\mu_1, \delta_2, \ldots, \delta_K, \phi, \eta^2, P\}. \tag{2.3.13}$$

Since P is row stochastic, it has $K(K-1)$ free parameters. Therefore, the total number of free parameters is $K + 2 + K(K-1) = K^2 + 2$.

Remark 2.3.1. In the case of $K = 1$, the MSSV model reduces to the standard stochastic volatility (SV) model, in which only the log volatility H_t is a latent variable and there are no regimes.

2.4 Problem formulation

The goal of this thesis is to estimate volatility in financial markets using the MSSV model with parameters θ. Let $y_{1:t} = \{y_1, y_2, \ldots, y_t\}$ denote the sequence of observed returns up to time $t \leq T$. Similarly, let $x_{0:t} = \{x_0, x_1, \ldots, x_t\}$ denote the sequence of latent states up to time t, where x_0 is the initial prior state. In general, we can distinguish between two versions of the problem.

1. **With known parameters.** If we assume that the model parameters θ are known, then the main objective is the smoothing distribution $p(x_{0:t} \mid y_{1:t}, \theta)$.

2. **With unknown parameters.** In reality, the parameters θ of the MSSV model are unknown, so we would also like to estimate them. The main objective is then the joint posterior distribution $p(x_{0:t}, \theta \mid y_{1:t})$,

from which we can compute the smoothing, filtering and predictive distributions of the latent states and the posterior distribution of the parameters, as in Definition 2.2.2.

Both targets are determined, up to normalizing constants, by the joint density of the latent states and observations.

Proposition 2.4.1. *For the MSSV model, the joint density of the latent states and observations is*

$$p(x_{0:t}, y_{1:t} \mid \theta) = \chi(x_0 \mid \theta) \prod_{i=1}^{t} p(x_i \mid x_{i-1}, \theta) f(y_i \mid x_i). \tag{2.4.1}$$

Consequently, with known parameters,

$$p(x_{0:t} \mid y_{1:t}, \theta) \propto \chi(x_0 \mid \theta) \prod_{i=1}^{t} p(x_i \mid x_{i-1}, \theta) f(y_i \mid x_i), \tag{2.4.2}$$

and, given a prior $p(\theta)$, with unknown parameters,

$$p(x_{0:t}, \theta \mid y_{1:t}) \propto p(\theta)\chi(x_0 \mid \theta) \prod_{i=1}^{t} p(x_i \mid x_{i-1}, \theta) f(y_i \mid x_i). \tag{2.4.3}$$

Proof. By the definition of conditional density,

$$p(x_{0:t}, y_{1:t} \mid \theta) = p(y_{1:t} \mid x_{0:t}, \theta)\, p(x_{0:t} \mid \theta). \tag{2.4.4}$$

The first factor can be decomposed as

$$\begin{aligned} p(y_{1:t} \mid x_{0:t}, \theta) &= p(y_t \mid y_{1:t-1}, x_{0:t}, \theta) p(y_{1:t-1} \mid x_{0:t}, \theta) && (2.4.5)\\ &= f(y_t \mid x_t, \theta) p(y_{1:t-1} \mid x_{0:t-1}, \theta) && (2.4.6)\\ &= \prod_{i=1}^{t} f(y_i \mid x_i), && (2.4.7) \end{aligned}$$

where we used the fact that each observation y_i is conditionally independent of the other observations and states given x_i (Definition 2.2.1).

Similarly, the second factor can be decomposed as

$$p(x_{0:t} \mid \theta) = \chi(x_0 \mid \theta) \prod_{i=1}^{t} p(x_i \mid x_{0:i-1}, \theta) = \chi(x_0 \mid \theta) \prod_{i=1}^{t} p(x_i \mid x_{i-1}, \theta). \tag{2.4.8}$$

Multiplying both factors yields (2.4.1).

For known parameters, Bayes' theorem gives

$$p(x_{0:t} \mid y_{1:t}, \theta) = \frac{p(x_{0:t}, y_{1:t} \mid \theta)}{p(y_{1:t} \mid \theta)}, \tag{2.4.9}$$

where the denominator does not depend on $x_{0:t}$, which yields (2.4.2). For unknown parameters,

$$p(x_{0:t}, \theta \mid y_{1:t}) = \frac{p(x_{0:t}, y_{1:t} \mid \theta)\, p(\theta)}{p(y_{1:t})}, \tag{2.4.10}$$

where the denominator does not depend on $(x_{0:t}, \theta)$, which yields (2.4.3). □

Equations (2.4.2) and (2.4.3) both reveal the sequential nature of the problem. If we initially sample the prior latent state x_0 (and the model parameters θ, if they are unknown), we can sequentially propagate the latent states through the transition density and update them with new observations through the observation density. This is the main idea behind sequential Monte Carlo methods, which are presented in the next chapter.

2.4.1 Prior distributions of the parameters

In the case of unknown model parameters θ, we need to specify their prior distributions. We assume that the priors of the parameters are independent, so that the joint prior distribution can be factorized as

$$p(\theta) = p(\mu_1) \left(\prod_{i=2}^{K} p(\delta_i) \right) p(\phi)\, p(\eta^2)\, p(P). \tag{2.4.11}$$

Even though the model is defined in terms of the parameters δ_i and ϕ, we define the priors in terms of $\Delta_i = \mu_i - \mu_{i-1} = e^{\delta_i}$ and $\varphi = \frac{\phi+1}{2}$. The reason for reparameterizing ϕ as φ is that φ lies in the range $(0, 1)$, which allows us to use the Beta distribution as a prior. The reason for reparameterizing δ_i as Δ_i is that we can then use a truncated normal distribution as a prior, which is more convenient for conditional sampling later on.

We choose the following prior distributions for the model parameters:

$$\mu_1 \sim \mathcal{N}(a_0, 5^2), \tag{2.4.12}$$

$$\Delta_i \sim \mathcal{TN}_{(0,\infty)}(0, 5^2), \qquad i = 2, \ldots, K, \tag{2.4.13}$$

$$\varphi \sim \mathcal{B}(6, 2), \tag{2.4.14}$$

$$\eta^2 \sim \mathcal{IG}(2.001, 0.1), \tag{2.4.15}$$

$$P_{i,:} \sim \mathcal{D}(\alpha_{i,1}, \ldots, \alpha_{i,K}), \qquad i = 1, \ldots, K, \tag{2.4.16}$$

where $\alpha_{i,j} = 1.5$ for $i \neq j$ and $\alpha_{i,i} = 40$. Here, $\mathcal{TN}_{(0,\infty)}$ denotes the truncated normal distribution with support $(0, \infty)$, $\mathcal{B}$ denotes the Beta distribution, $\mathcal{IG}$ denotes the inverse gamma distribution, and $\mathcal{D}$ denotes the Dirichlet distribution.

The priors are chosen to be informative while allowing sufficient flexibility. For μ_1, a_0 is determined through prior elicitation, while the variance 5^2 allows for substantial uncertainty. The prior for Δ_i allows for a wide range of possible values, but still encourages the regime means not to be too far apart. The $\mathcal{B}(6, 2)$ prior for φ favors positive persistence, as is common in financial econometrics. Additionally, previous results from [13] suggest that the choice of prior for this parameter has little effect on the posterior, so we chose hyperparameters that lie in between the values considered there. For η^2, we use $\mathcal{IG}(2.001, 0.1)$, following a combination of the values considered in [10, 13]. This prior has a large variance, allowing for a wide range of values. For the transition matrix P, we depart from the commonly used symmetric Dirichlet prior with hyperparameters 0.5 [9, 10, 13]. Such a prior places substantial mass near 0 and 1, which may encourage regimes to either disappear or become excessively persistent. Instead, motivated by [14], we use $\alpha_{i,i} = 40$ and $\alpha_{i,j} = 1.5$ for $i \neq j$. In the two-regime case, this corresponds to a $\mathcal{B}(40, 1.5)$ prior for the diagonal transition probabilities, with mean 0.964, implying an expected regime duration of approximately 28 days. We find this expected duration reasonable for financial data.

Chapter 3

Sequential Monte Carlo Methods

Sequential Monte Carlo (SMC) methods provide a flexible framework for approximating probability distributions that evolve sequentially over time. They are particularly useful for inference in state-space models, where the latent states are typically not directly observed and their posterior distributions are analytically intractable. SMC methods approximate these distributions using a set of weighted samples, commonly referred to as particles, and update this approximation as new observations become available.

This chapter first introduces the foundations of SMC, beginning with Monte Carlo methods, importance sampling and sequential importance sampling. The problem of weight degeneracy is then addressed through resampling, leading to sequential importance sampling with resampling, also known as particle filtering. The bootstrap particle filter is presented in more detail.

Next, the chapter considers particle-based methods for Bayesian parameter inference. Particle marginal Metropolis-Hastings and particle Gibbs are introduced as particle Markov chain Monte Carlo methods. Finally, SMC^2 is presented as an extension that combines particle filtering with sequential inference for static parameters and latent states.

3.1 Monte Carlo methods

Monte Carlo methods approximate quantities involving a probability distribution by using a finite collection of random samples. The basic idea is simple: rather than evaluating an integral directly, we replace it by an average over samples drawn from the distribution of interest.

Until further notice, let X be a random variable with probability density function $p(x)$, and let g be a measurable function such that $E_p[|g(X)|] < \infty$. The quantity of interest is

$$I = E_p[g(X)] = \int g(x)p(x)\,\mathrm{d}x. \tag{3.1.1}$$

Definition 3.1.1. Let $X_1, \ldots, X_N \overset{\text{iid}}{\sim} p(x)$. The *Monte Carlo estimator* of I is defined as

$$\hat{I}_N = \frac{1}{N}\sum_{i=1}^{N} g(X_i). \tag{3.1.2}$$

Such an estimator has several desirable properties, including unbiasedness, consistency, and asymptotic normality.

Proposition 3.1.1. *The Monte Carlo estimator $\hat{I}_N$ is unbiased.*

Proof. The proof follows from the linearity of expectation and the fact that the samples are identically distributed.

$$E_p[\hat{I}_N] = E_p\left[\frac{1}{N}\sum_{i=1}^{N} g(X_i)\right] \tag{3.1.3}$$

$$= \frac{1}{N}\sum_{i=1}^{N} E_p[g(X_i)] \tag{3.1.4}$$

$$= \frac{1}{N}\sum_{i=1}^{N} I = I. \tag{3.1.5}$$

□

Proposition 3.1.2. *The Monte Carlo estimator $\hat{I}_N$ is consistent.*

Proof. Because the samples are i.i.d. and $E_p[|g(X)|] < \infty$, the strong law of large numbers applies, giving

$$\hat{I}_N \xrightarrow[N\to\infty]{a.s.} I. \tag{3.1.6}$$

Almost sure convergence implies convergence in probability, so the estimator is consistent. □

Proposition 3.1.3. *If $g(X)$ has finite variance, then the Monte Carlo estimator $\hat{I}_N$ is asymptotically normal.*

Proof. If $g(X)$ has finite variance σ^2, then the central limit theorem applies, giving

$$\sqrt{N}\left(\hat{I}_N - I\right) \xrightarrow[N\to\infty]{D} \mathcal{N}(0, \sigma^2), \tag{3.1.7}$$

which is the definition of asymptotic normality. □

3.1.1 Empirical distribution

The Monte Carlo estimator can be interpreted as integration with respect to an empirical approximation constructed from the samples.

Definition 3.1.2. Given samples $X_1, \ldots, X_N \overset{\text{iid}}{\sim} p(x)$, the *empirical distribution* associated with these samples is defined as

$$\hat{p}_N(x) = \frac{1}{N}\sum_{i=1}^{N} \delta(x - X_i), \tag{3.1.8}$$

where $\delta(\cdot)$ denotes the Dirac delta function.

The empirical distribution assigns equal weight $1/N$ to each sampled value. It therefore provides a discrete approximation to the distribution represented by $p(x)$. The expression in (3.1.8) should be understood through integration rather than as an ordinary probability density with respect to Lebesgue measure.

Proposition 3.1.4. *For g, $\hat{I}_N$ and $\hat{p}_N$ as previously defined,*

$$\hat{I}_N = \int g(x)\hat{p}_N(x)\,\mathrm{d}x. \tag{3.1.9}$$

Proof.

$$\int g(x)\hat{p}_N(x)\,\mathrm{d}x = \int g(x)\frac{1}{N}\sum_{i=1}^{N}\delta(x - X_i)\,\mathrm{d}x \tag{3.1.10}$$

$$= \frac{1}{N}\sum_{i=1}^{N}\int g(x)\delta(x - X_i)\,\mathrm{d}x \tag{3.1.11}$$

$$= \frac{1}{N}\sum_{i=1}^{N} g(X_i) = \hat{I}_N, \tag{3.1.12}$$

where the defining property of the Dirac delta function has been used. □

It is important to distinguish the empirical distribution $\hat{p}_N$ from the density p. Even when p is continuous, $\hat{p}_N$ consists of point masses for every finite N and is therefore not a pointwise approximation of $p(x)$. The approximation is instead meaningful through quantities such as integrals and expectations.

3.2 Importance sampling

The standard Monte Carlo method requires samples from the target distribution p. In many problems, however, sampling directly from p is difficult or even impossible. Importance sampling addresses this problem by introducing a second distribution from which sampling is easier.

Definition 3.2.1. Let $q(x)$ be a proposal distribution satisfying $q(x) > 0$ whenever $p(x) > 0$. The *importance weight* associated with a point x is defined as

$$w(x) = \frac{p(x)}{q(x)}. \tag{3.2.1}$$

The expectation of g under p can then be rewritten as

$$I = E_p[g(X)] = \int g(x)p(x)\,\mathrm{d}x \tag{3.2.2}$$

$$= \int g(x)\frac{p(x)}{q(x)}q(x)\,\mathrm{d}x \tag{3.2.3}$$

$$= E_q\left[g(X)w(X)\right]. \tag{3.2.4}$$

Thus, although samples are generated from q, the importance weights correct for the difference between q and p.

Definition 3.2.2. Let $X_1, \ldots, X_N \overset{\text{iid}}{\sim} q$ and $w(X_1), \ldots, w(X_N)$ their corresponding importance weights. The *importance sampling estimator* of $I = E_p[g(X)]$ is defined by

$$\hat{I}_N^{\text{IS}} = \frac{1}{N} \sum_{i=1}^{N} g(X_i) w(X_i). \tag{3.2.5}$$

Proposition 3.2.1. *The importance sampling estimator $\hat{I}_N^{\text{IS}}$ is unbiased.*

Proof.

$$E_q[\hat{I}_N^{\text{IS}}] = \frac{1}{N} \sum_{i=1}^{N} E_q[g(X_i) w(X_i)] \tag{3.2.6}$$

$$= E_q \left[g(X) \frac{p(X)}{q(X)} \right] \tag{3.2.7}$$

$$= \int g(x) \frac{p(x)}{q(x)} q(x) \, \mathrm{d}x = I. \tag{3.2.8}$$

□

Proposition 3.2.2. *The importance sampling estimator $\hat{I}_N^{\text{IS}}$ is consistent.*

Proof. Because $E_q[|g(X)w(X)|] = E_p[|g(X)|] < \infty$, the strong law of large numbers gives

$$\hat{I}_N^{\text{IS}} \xrightarrow[N \to \infty]{a.s.} I, \tag{3.2.9}$$

which means that the estimator is consistent. □

Although the importance sampling estimator is unbiased and consistent for any proposal distribution satisfying the support condition, its variance depends strongly on the choice of q.

3.2.1 (Weighted) empirical distribution

There are two equivalent ways to interpret the importance sampling estimator. The first one follows the same logic as in Section 3.1.1:

$$\hat{I}_N^{\mathrm{IS}} = \frac{1}{N}\sum_{i=1}^{N} w(X_i)g(X_i) = \frac{1}{N}\sum_{i=1}^{N}\int w(x)g(x)\delta(x - X_i)\,\mathrm{d}x \tag{3.2.10}$$

$$= \int w(x)g(x)\frac{1}{N}\sum_{i=1}^{N}\delta(x - X_i)\,\mathrm{d}x \tag{3.2.11}$$

$$= \int w(x)g(x)\hat{q}_N(x)\,\mathrm{d}x, \tag{3.2.12}$$

where $\hat{q}_N$ is the empirical distribution of the samples $X_1, \ldots, X_N \overset{\text{iid}}{\sim} q$, so the estimator is simply the expectation of $w(X)g(X)$ with respect to $\hat{q}_N$.

There is another interpretation that is particularly useful. Instead of viewing the weights as part of the integrand, we can absorb them into the empirical distribution itself.

$$\hat{I}_N^{\mathrm{IS}} = \frac{1}{N}\sum_{i=1}^{N} w(X_i)g(X_i) = \frac{1}{N}\sum_{i=1}^{N} w(X_i)\int g(x)\delta(x - X_i)\,\mathrm{d}x \tag{3.2.13}$$

$$= \int g(x)\frac{1}{N}\sum_{i=1}^{N} w(X_i)\delta(x - X_i)\,\mathrm{d}x \tag{3.2.14}$$

$$= \int g(x)\hat{p}_N^{\mathrm{IS}}(x)\,\mathrm{d}x, \tag{3.2.15}$$

leading us to next definition.

Definition 3.2.3. Let $X_1, \ldots, X_N \overset{\text{iid}}{\sim} q$ and $w(X_1), \ldots, w(X_N)$ their corresponding importance weights. The *weighted empirical distribution* is defined as

$$\hat{p}_N^{\mathrm{IS}}(x) = \frac{1}{N}\sum_{i=1}^{N} w(X_i)\delta(x - X_i). \tag{3.2.16}$$

This provides an alternative interpretation of importance sampling: although the samples are drawn from the proposal distribution q, the importance weights allow the samples to represent the target distribution p. Unlike

ordinary Monte Carlo, where every sample contributes the same mass $1/N$, importance sampling assigns the sample X_i a mass proportional to $w(X_i)$.

The weighted empirical distribution in (3.2.16), however, is not generally a probability distribution for a finite sample. Its total mass is

$$\int \hat{p}_N^{\mathrm{IS}}(x)\,\mathrm{d}x = \frac{1}{N}\sum_{i=1}^{N} w(X_i), \tag{3.2.17}$$

which is a random quantity and is not necessarily equal to one. Nevertheless, the total mass has expectation equal to one. Since $X \sim q$,

$$E_q[w(X)] = \int \frac{p(x)}{q(x)} q(x)\,\mathrm{d}x = \int p(x)\,\mathrm{d}x = 1, \tag{3.2.18}$$

$$E_q\left[\frac{1}{N}\sum_{i=1}^{N} w(X_i)\right] = \frac{1}{N}\sum_{i=1}^{N} E_q[w(X_i)] = 1. \tag{3.2.19}$$

Additionally, the strong law of large numbers gives

$$\frac{1}{N}\sum_{i=1}^{N} w(X_i) \xrightarrow[N\to\infty]{a.s.} E_q[w(X)] = 1. \tag{3.2.20}$$

Thus, although the weighted empirical distribution does not necessarily have unit mass for a finite sample, its total mass converges almost surely to one as the number of samples increases.

3.3 Self-normalized importance sampling

The previous construction suggests a natural modification: instead of allowing the weights to have a random total mass, normalize them. Writing $w_i = w(X_i)$ for brevity, the *normalized importance weights* are

$$\tilde{w}_i = \frac{w_i}{\sum_{j=1}^{N} w_j}, \tag{3.3.1}$$

which leads to the following definition.

Definition 3.3.1. Let $X_1, \ldots, X_N \overset{\text{iid}}{\sim} q$ and $\tilde{w}_1, \ldots, \tilde{w}_N$ their corresponding normalized importance weights. The *self-normalized weighted empirical distribution* is defined as

$$\hat{p}_N^{\text{SNIS}}(x) = \sum_{i=1}^{N} \tilde{w}_i \delta(x - X_i). \tag{3.3.2}$$

This distribution clearly has a total mass of one.

Definition 3.3.2. Let $X_1, \ldots, X_N \overset{\text{iid}}{\sim} q$ and $w_1, \ldots, w_N$ their corresponding importance weights. The *self-normalized importance sampling estimator* of $I = E_p[g(X)]$ is defined as

$$\hat{I}_N^{\text{SNIS}} = \frac{\sum_{i=1}^{N} g(X_i) w_i}{\sum_{i=1}^{N} w_i}. \tag{3.3.3}$$

Proposition 3.3.1. *For g, $\hat{I}_N^{\text{SNIS}}$ and $\hat{p}_N^{\text{SNIS}}$ as previously defined,*

$$\hat{I}_N^{\text{SNIS}} = \int g(x) \hat{p}_N^{\text{SNIS}}(x) \, \mathrm{d}x. \tag{3.3.4}$$

Proof.

$$\int g(x) \hat{p}_N^{\text{SNIS}}(x) \, \mathrm{d}x = \int g(x) \sum_{i=1}^{N} \tilde{w}_i \delta(x - X_i) \, \mathrm{d}x \tag{3.3.5}$$

$$= \sum_{i=1}^{N} \tilde{w}_i \int g(x) \delta(x - X_i) \, \mathrm{d}x \tag{3.3.6}$$

$$= \sum_{i=1}^{N} \tilde{w}_i g(X_i) \tag{3.3.7}$$

$$= \frac{\sum_{i=1}^{N} g(X_i) w_i}{\sum_{i=1}^{N} w_i}. \tag{3.3.8}$$

□

There is a fundamental difference between $\hat{I}_N^{\text{IS}}$ and $\hat{I}_N^{\text{SNIS}}$. The ordinary importance sampling estimator is unbiased, whereas the self-normalized estimator is generally biased for finite N.

Proposition 3.3.2. *The self-normalized importance sampling estimator is, in general, biased.*

Proof. In general, the expected value of a ratio is not equal to the ratio of expected values, so

$$E_q\left[\frac{\sum_{i=1}^N g(X_i)w_i}{\sum_{i=1}^N w_i}\right] \neq \frac{E_q\left[\frac{1}{N}\sum_{i=1}^N g(X_i)w_i\right]}{E_q\left[\frac{1}{N}\sum_{i=1}^N w_i\right]}, \tag{3.3.9}$$

$$E_q\left[\hat{I}_N^{\text{SNIS}}\right] \neq \frac{E_q\left[\hat{I}_N^{\text{IS}}\right]}{1} = I. \tag{3.3.10}$$

□

Proposition 3.3.3. *The self-normalized importance sampling estimator is consistent.*

Proof. The strong law of large numbers gives

$$\frac{1}{N}\sum_{i=1}^N g(X_i)w_i \xrightarrow[N\to\infty]{a.s.} I, \tag{3.3.11}$$

$$\frac{1}{N}\sum_{i=1}^N w_i \xrightarrow[N\to\infty]{a.s.} 1. \tag{3.3.12}$$

Consequently,

$$\hat{I}_N^{\text{SNIS}} = \frac{\frac{1}{N}\sum_{i=1}^N g(X_i)w_i}{\frac{1}{N}\sum_{i=1}^N w_i} \xrightarrow[N\to\infty]{a.s.} I. \tag{3.3.13}$$

□

At first, it may seem pointless to use the self-normalized estimator merely to ensure that $\hat{p}_N^{\text{SNIS}}$ is a proper probability distribution, since the estimator loses its unbiasedness. However, in many applications, especially in Bayesian inference, the target density $p(x)$ is known only up to a proportionality constant. Suppose that the target has the form

$$p(x) = \frac{\tilde{p}(x)}{Z}, \tag{3.3.14}$$

where $\tilde{p}(x)$ is known and Z is an intractable normalization constant. The importance weights are then

$$w(x) = \frac{p(x)}{q(x)} = \frac{\tilde{p}(x)}{Zq(x)}, \tag{3.3.15}$$

which makes it impossible to compute the ordinary importance sampling estimator $\hat{I}_N^{\mathrm{IS}}$. However, the normalized weights are still computable:

$$\tilde{w}_i = \frac{w_i}{\sum_{j=1}^N w_j} = \frac{\frac{\tilde{p}(X_i)}{Zq(X_i)}}{\sum_{j=1}^N \frac{\tilde{p}(X_j)}{Zq(X_j)}} = \frac{\frac{\tilde{p}(X_i)}{q(X_i)}}{\sum_{j=1}^N \frac{\tilde{p}(X_j)}{q(X_j)}}. \tag{3.3.16}$$

Therefore, the self-normalized importance sampling estimator $\hat{I}_N^{\mathrm{SNIS}}$ can be used even when the target density is known only up to a constant of proportionality, at the cost of introducing bias.

3.4 Sequential importance sampling

Sequential importance sampling (SIS) applies importance sampling to a sequence of random variables by exploiting the sequential structure of the problem. Because the main focus of this thesis is on state-space models, in particular the MSSV model, we consider sequential importance sampling in the context of state-space models.

For the moment, assume that the model parameters θ are known. Given the observations $y_{1:T}$, the objective is to approximate the posterior distribution of the latent states, $p(x_{0:T} \mid y_{1:T}, \theta)$. By Proposition 2.4.1, the joint density is

$$p(x_{0:T}, y_{1:T} \mid \theta) = \chi(x_0 \mid \theta) \prod_{t=1}^{T} f(y_t \mid x_t, \theta) p(x_t \mid x_{t-1}, \theta). \tag{3.4.1}$$

For the MSSV model, the latent state $x_t = (h_t, s_t)$ has a continuous component h_t and a discrete component $s_t \in \{1, \ldots, K\}$. All densities of the latent states are therefore understood with respect to the product of the Lebesgue

measure and the counting measure, so that integrals over x_t are shorthand for

$$\int g(x_t)\,\mathrm{d}x_t = \sum_{s_t=1}^{K} \int g(h_t, s_t)\,\mathrm{d}h_t. \tag{3.4.2}$$

Correspondingly, $\delta(x_t - X_t)$ acts as a Dirac delta in h_t and as a Kronecker delta in s_t. With this convention, all results of this chapter apply to the MSSV model without change.

Until further notice, assume that the proposal distribution can also be factorized sequentially as

$$q(x_{0:T} \mid y_{1:T}, \theta) = q_0(x_0 \mid \theta) \prod_{t=1}^{T} q_t(x_t \mid x_{0:t-1}, y_{1:t}, \theta), \tag{3.4.3}$$

from which we can sample entire trajectories of the latent states, conditioned on the currently available observations, in a sequential manner.

The importance weight can therefore be computed recursively as

$$\begin{aligned} w_T &= \frac{p(x_{0:T}, y_{1:T} \mid \theta)}{q(x_{0:T} \mid y_{1:T}, \theta)} && (3.4.4)\\ &= \frac{\chi(x_0 \mid \theta) \prod_{t=1}^{T} f(y_t \mid x_t, \theta) p(x_t \mid x_{t-1}, \theta)}{q_0(x_0 \mid \theta) \prod_{t=1}^{T} q_t(x_t \mid x_{0:t-1}, y_{1:t}, \theta)} && (3.4.5)\\ &= w_{T-1} \cdot \frac{f(y_T \mid x_T, \theta) p(x_T \mid x_{T-1}, \theta)}{q_T(x_T \mid x_{0:T-1}, y_{1:T}, \theta)} && (3.4.6)\\ &= w_{T-1} \alpha_T = \prod_{t=0}^{T} \alpha_t && (3.4.7) \end{aligned}$$

with $\alpha_0 = w_0 = \frac{\chi(x_0|\theta)}{q_0(x_0|\theta)}$.

Definition 3.4.1. The *marginal likelihood* is defined as

$$p(y_{1:T} \mid \theta) = \int p(x_{0:T}, y_{1:T} \mid \theta)\,\mathrm{d}x_{0:T}. \tag{3.4.8}$$

Introducing the importance weights, the marginal likelihood can therefore be written as

$$p(y_{1:T} \mid \theta) = \int w_T\, q(x_{0:T} \mid y_{1:T}, \theta)\,\mathrm{d}x_{0:T} = E_q[w_T]. \tag{3.4.9}$$

For nonlinear and non-Gaussian state-space models, the marginal likelihood is typically intractable and therefore has to be estimated.

Definition 3.4.2. Let $X_{0:T}^{(1)}, \ldots, X_{0:T}^{(N)} \overset{\text{iid}}{\sim} q$ be sampled trajectories and $w_T^{(i)}$ their corresponding importance weights. Then the *marginal likelihood estimator* is defined as

$$\hat{p}(y_{1:T} \mid \theta) = \frac{1}{N} \sum_{i=1}^{N} w_T^{(i)}. \tag{3.4.10}$$

As a result the marginal likelihood estimator can therefore be written as

$$\hat{p}(y_{1:T} \mid \theta) = \frac{1}{N} \sum_{i=1}^{N} \prod_{t=0}^{T} \alpha_t^{(i)}. \tag{3.4.11}$$

Proposition 3.4.1. *The marginal likelihood estimator $\hat{p}(y_{1:T} \mid \theta)$ is unbiased and consistent.*

Proof. Unbiasedness follows from

$$E_q[\hat{p}(y_{1:T} \mid \theta)] = E_q\left[\frac{1}{N} \sum_{i=1}^{N} w_T^{(i)}\right] = \frac{1}{N} \sum_{i=1}^{N} E_q[w_T^{(i)}] \tag{3.4.12}$$

$$= E_q[w_T] = \int w_T q(x_{0:T} \mid y_{1:T}, \theta) \, \mathrm{d}x_{0:T} \tag{3.4.13}$$

$$= \int p(x_{0:T}, y_{1:T} \mid \theta) \, \mathrm{d}x_{0:T} = p(y_{1:T} \mid \theta), \tag{3.4.14}$$

where the linearity of expectation and the fact that the trajectories are identically distributed have been used. Since the weights $w_T^{(i)}$ are i.i.d., non-negative and have finite expectation $p(y_{1:T} \mid \theta)$, consistency follows from the strong law of large numbers,

$$\hat{p}(y_{1:T} \mid \theta) \xrightarrow[N \to \infty]{a.s.} p(y_{1:T} \mid \theta). \tag{3.4.15}$$

Alternatively, $\hat{p}(y_{1:T} \mid \theta)$ is the (ordinary) importance sampling estimator $\hat{I}_N^{\text{IS}}$ with $g \equiv 1$, hence the proofs of Propositions 3.2.1 and 3.2.2 carry over directly. □

3.4.1 Smoothing distribution

Definition 3.4.3. The *smoothing distribution* is the posterior distribution of the entire latent trajectory given all available observations,

$$p(x_{0:T} \mid y_{1:T}, \theta) = \frac{p(x_{0:T}, y_{1:T} \mid \theta)}{p(y_{1:T} \mid \theta)}. \tag{3.4.16}$$

The *smoothing expectation* of a function g is defined as

$$I^{\text{smth}} = E\left[g(X_{0:T}) \mid y_{1:T}, \theta\right] = \int g(x_{0:T}) p(x_{0:T} \mid y_{1:T}, \theta) \, \mathrm{d}x_{0:T}. \tag{3.4.17}$$

Rewriting the smoothing expectation in terms of the proposal distribution gives

$$I^{\text{smth}} = \frac{1}{p(y_{1:T} \mid \theta)} \int g(x_{0:T}) \frac{p(x_{0:T}, y_{1:T} \mid \theta)}{q(x_{0:T} \mid y_{1:T}, \theta)} q(x_{0:T} \mid y_{1:T}, \theta) \, \mathrm{d}x_{0:T} \tag{3.4.18}$$

$$= \frac{E_q\left[g(X_{0:T}) w_T\right]}{p(y_{1:T} \mid \theta)} = \frac{E_q\left[g(X_{0:T}) w_T\right]}{E_q\left[w_T\right]}. \tag{3.4.19}$$

The smoothing distribution is thus a target known only up to the normalizing constant $p(y_{1:T} \mid \theta)$, with unnormalized density $p(x_{0:T}, y_{1:T} \mid \theta)$, which is exactly the setting of Section 3.3. Replacing both expectations by Monte Carlo averages over the sampled trajectories motivates the following definitions.

Definition 3.4.4. Let $X_{0:T}^{(i)} \overset{\text{iid}}{\sim} q$ be sampled trajectories, $w_T^{(i)}$ their importance weights, and

$$\tilde{w}_T^{(i)} = \frac{w_T^{(i)}}{\sum_{j=1}^N w_T^{(j)}} \tag{3.4.20}$$

the corresponding normalized weights. The *smoothing estimator* of I^{smth} is defined as

$$\hat{I}_N^{\text{smth}} = \frac{\sum_{i=1}^N g(X_{0:T}^{(i)}) w_T^{(i)}}{\sum_{i=1}^N w_T^{(i)}} = \sum_{i=1}^N \tilde{w}_T^{(i)} g(X_{0:T}^{(i)}), \tag{3.4.21}$$

inducing the *empirical smoothing distribution* as

$$\hat{p}_N^{\text{smth}}(x_{0:T} \mid y_{1:T}, \theta) = \sum_{i=1}^N \tilde{w}_T^{(i)} \delta(x_{0:T} - X_{0:T}^{(i)}). \tag{3.4.22}$$

Proposition 3.4.2. *For g, $\hat{I}_N^{\text{smth}}$ and $\hat{p}_N^{\text{smth}}$ as previously defined,*

$$\hat{I}_N^{\text{smth}} = \int g(x_{0:T}) \hat{p}_N(x_{0:T} \mid y_{1:T}, \theta) \, \mathrm{d}x_{0:T}. \tag{3.4.23}$$

Moreover, the smoothing estimator $\hat{I}_N^{\text{smth}}$ is, in general, biased, but consistent.

Proof. The first claim follows from the defining property of the Dirac delta function,

$$\int g(x_{0:T})\hat{p}_N(x_{0:T} \mid y_{1:T}, \theta)\,\mathrm{d}x_{0:T} = \sum_{i=1}^{N} \tilde{w}_T^{(i)} \int g(x_{0:T})\delta(x_{0:T} - X_{0:T}^{(i)})\,\mathrm{d}x_{0:T} \tag{3.4.24}$$

$$= \sum_{i=1}^{N} \tilde{w}_T^{(i)} g(X_{0:T}^{(i)}) = \hat{I}_N^{\mathrm{smth}}. \tag{3.4.25}$$

The estimator $\hat{I}_N^{\mathrm{smth}}$ is the self-normalized importance sampling estimator $\hat{I}_N^{\mathrm{SNIS}}$ with target $p(x_{0:T} \mid y_{1:T}, \theta)$, unnormalized target $p(x_{0:T}, y_{1:T} \mid \theta)$, normalizing constant $Z = p(y_{1:T} \mid \theta)$ and proposal $q(x_{0:T} \mid y_{1:T}, \theta)$. Bias and consistency therefore follow from Propositions 3.3.2 and 3.3.3. □

3.4.2 Filtering distribution

Definition 3.4.5. The *filtering distribution* is the marginal of the smoothing distribution,

$$p(x_T \mid y_{1:T}, \theta) = \int p(x_{0:T} \mid y_{1:T}, \theta)\,\mathrm{d}x_{0:T-1}. \tag{3.4.26}$$

The *filtering expectation* of a function g is defined as

$$I^{\mathrm{filt}} = E\left[g(X_T) \mid y_{1:T}, \theta\right] = \int g(x_T)p(x_T \mid y_{1:T}, \theta)\,\mathrm{d}x_T. \tag{3.4.27}$$

In the empirical setting, the states at previous times are easily integrated out, since $\delta(x_{0:T} - X_{0:T}^{(i)}) = \delta(x_T - X_T^{(i)})\,\delta(x_{0:T-1} - X_{0:T-1}^{(i)})$,

$$\int \hat{p}_N(x_{0:T} \mid y_{1:T}, \theta)\,\mathrm{d}x_{0:T-1} \tag{3.4.28}$$

$$= \sum_{i=1}^{N} \tilde{w}_T^{(i)} \delta(x_T - X_T^{(i)}) \int \delta(x_{0:T-1} - X_{0:T-1}^{(i)})\,\mathrm{d}x_{0:T-1} \tag{3.4.29}$$

$$= \sum_{i=1}^{N} \tilde{w}_T^{(i)} \delta(x_T - X_T^{(i)}), \tag{3.4.30}$$

which leads to the following definition.

Definition 3.4.6. Let $X_{0:T}^{(i)} \overset{\text{iid}}{\sim} q$ and $\tilde{w}_T^{(i)}$ their corresponding normalized weights. The *filtering estimator* of I^{filt} is defined as

$$\hat{I}_N^{\text{filt}} = \sum_{i=1}^{N} \tilde{w}_T^{(i)} g(X_T^{(i)}), \tag{3.4.31}$$

inducing the *empirical filtering distribution* is as

$$\hat{p}_N^{\text{filt}}(x_T \mid y_{1:T}, \theta) = \sum_{i=1}^{N} \tilde{w}_T^{(i)} \delta(x_T - X_T^{(i)}). \tag{3.4.32}$$

Proposition 3.4.3. *The filtering estimator $\hat{I}_N^{\text{filt}}$ is, in general, biased, but consistent.*

Proof. Define the function $\bar{g}(x_{0:T}) = g(x_T)$, which depends on the trajectory only through its final state. Its smoothing expectation is

$$E\left[\bar{g}(X_{0:T}) \mid y_{1:T}, \theta\right] = \int g(x_T)\, p(x_{0:T} \mid y_{1:T}, \theta)\, \mathrm{d}x_{0:T} \tag{3.4.33}$$

$$= \int g(x_T) \left(\int p(x_{0:T} \mid y_{1:T}, \theta)\, \mathrm{d}x_{0:T-1} \right) \mathrm{d}x_T \tag{3.4.34}$$

$$= \int g(x_T)\, p(x_T \mid y_{1:T}, \theta)\, \mathrm{d}x_T = I^{\text{filt}}, \tag{3.4.35}$$

where the previous states have been integrated out. Likewise, the smoothing estimator of this expectation is

$$\sum_{i=1}^{N} \tilde{w}_T^{(i)} \bar{g}(X_{0:T}^{(i)}) = \sum_{i=1}^{N} \tilde{w}_T^{(i)} g(X_T^{(i)}) = \hat{I}_N^{\text{filt}}. \tag{3.4.36}$$

Thus, $\hat{I}_N^{\text{filt}}$ is the smoothing estimator $\hat{I}_N^{\text{smth}}$ applied to $\bar{g}$, and its bias and consistency follow from Proposition 3.4.2. □

3.4.3 Predictive distribution

Definition 3.4.7. The (one-step) *predictive distribution* is obtained by propagating the filtering distribution through the transition density,

$$p(x_{T+1} \mid y_{1:T}, \theta) = \int p(x_{T+1} \mid x_T, \theta) p(x_T \mid y_{1:T}, \theta)\, \mathrm{d}x_T. \tag{3.4.37}$$

The *predictive expectation* of a function g is defined as

$$I^{\text{pred}} = E\left[g(X_{T+1}) \mid y_{1:T}, \theta\right] = \int g(x_{T+1}) p(x_{T+1} \mid y_{1:T}, \theta)\, \mathrm{d}x_{T+1}. \quad (3.4.38)$$

Replacing the filtering distribution with its empirical approximation gives

$$\int p(x_{T+1} \mid x_T, \theta) \hat{p}_N(x_T \mid y_{1:T}, \theta)\, \mathrm{d}x_T = \sum_{i=1}^{N} \tilde{w}_T^{(i)} p(x_{T+1} \mid X_T^{(i)}, \theta), \quad (3.4.39)$$

a mixture of transition densities weighted by the normalized weights. A particle approximation of this mixture is obtained by propagating each particle through its transition density, which leads to the following definition.

Definition 3.4.8. Let $X_{0:T}^{(i)}$ and $\tilde{w}_T^{(i)}$ be as in Definition 3.4.4, and let $X_{T+1}^{(i)} \sim p(x_{T+1} \mid X_T^{(i)}, \theta)$ be sampled independently. The *predictive estimator* of I^{pred} is defined as

$$\hat{I}_N^{\text{pred}} = \sum_{i=1}^{N} \tilde{w}_T^{(i)} g(X_{T+1}^{(i)}), \quad (3.4.40)$$

inducing the *empirical predictive distribution* is as

$$\hat{p}_N(x_{T+1} \mid y_{1:T}, \theta) = \sum_{i=1}^{N} \tilde{w}_T^{(i)} \delta(x_{T+1} - X_{T+1}^{(i)}). \quad (3.4.41)$$

Proposition 3.4.4. *The predictive estimator $\hat{I}_N^{\text{pred}}$ is, in general, biased, but consistent.*

Proof. The extended trajectories $X_{0:T+1}^{(i)}$ are independent draws from the extended proposal

$$\bar{q}(x_{0:T+1} \mid y_{1:T}, \theta) = q(x_{0:T} \mid y_{1:T}, \theta)\, p(x_{T+1} \mid x_T, \theta). \quad (3.4.42)$$

Take as the target the extended smoothing distribution

$$p(x_{0:T+1} \mid y_{1:T}, \theta) = \frac{p(x_{0:T}, y_{1:T} \mid \theta)\, p(x_{T+1} \mid x_T, \theta)}{p(y_{1:T} \mid \theta)}, \quad (3.4.43)$$

whose marginal in x_{T+1} is the predictive distribution, so that

$$I^{\text{pred}} = \int g(x_{T+1})\, p(x_{0:T+1} \mid y_{1:T}, \theta)\, \mathrm{d}x_{0:T+1}. \quad (3.4.44)$$

The transition densities cancel in the corresponding unnormalized importance weight,

$$\bar{w}_{T+1} = \frac{p(x_{0:T}, y_{1:T} \mid \theta)\, p(x_{T+1} \mid x_T, \theta)}{q(x_{0:T} \mid y_{1:T}, \theta)\, p(x_{T+1} \mid x_T, \theta)} = w_T. \tag{3.4.45}$$

Rewriting the predictive expectation in terms of the extended proposal therefore gives

$$I^{\text{pred}} = \frac{1}{p(y_{1:T} \mid \theta)} \int g(x_{T+1})\, \bar{w}_{T+1}\, \bar{q}(x_{0:T+1} \mid y_{1:T}, \theta)\, \mathrm{d}x_{0:T+1} \tag{3.4.46}$$

$$= \frac{E_{\bar{q}}\left[g(X_{T+1})\, w_T\right]}{p(y_{1:T} \mid \theta)}. \tag{3.4.47}$$

Analogously, the normalizing constant is

$$p(y_{1:T} \mid \theta) = \int \bar{w}_{T+1}\, \bar{q}(x_{0:T+1} \mid y_{1:T}, \theta)\, \mathrm{d}x_{0:T+1} = E_{\bar{q}}\left[w_T\right], \tag{3.4.48}$$

which yields

$$I^{\text{pred}} = \frac{E_{\bar{q}}\left[g(X_{T+1})\, w_T\right]}{E_{\bar{q}}\left[w_T\right]}. \tag{3.4.49}$$

Replacing both expectations by Monte Carlo averages over $X_{0:T+1}^{(i)} \overset{\text{iid}}{\sim} \bar{q}$ gives

$$\frac{\frac{1}{N}\sum_{i=1}^{N} g(X_{T+1}^{(i)})\, w_T^{(i)}}{\frac{1}{N}\sum_{i=1}^{N} w_T^{(i)}} = \sum_{i=1}^{N} \tilde{w}_T^{(i)} g(X_{T+1}^{(i)}) = \hat{I}_N^{\text{pred}}. \tag{3.4.50}$$

Hence $\hat{I}_N^{\text{pred}}$ is exactly the self-normalized importance sampling estimator $\hat{I}_N^{\text{SNIS}}$ of I^{pred} with proposal $\bar{q}$, unnormalized target $p(x_{0:T}, y_{1:T} \mid \theta)\, p(x_{T+1} \mid x_T, \theta)$ and normalizing constant $Z = p(y_{1:T} \mid \theta)$. Bias and consistency follow from Propositions 3.3.2 and 3.3.3. □

Algorithm 3.4.1 Sequential importance sampling

Require: Observations $y_{1:T}$, number of particles N, proposal densities q_0, q_t, and model densities χ, p, f.

1: **for** $i = 1, \ldots, N$ **do**
2: Sample $X_0^{(i)} \sim q_0(x_0 \mid \theta)$.
3: Set $w_0^{(i)} = \dfrac{\chi(X_0^{(i)} \mid \theta)}{q_0(X_0^{(i)} \mid \theta)}$.
4: **end for**
5: Normalize weights to obtain $\tilde{w}_0^{(i)}$.
6: **for** $t = 1, \ldots, T$ **do**
7: **for** $i = 1, \ldots, N$ **do**
8: Propagate $X_t^{(i)} \sim q_t(\cdot \mid X_{0:t-1}^{(i)}, y_{1:t}, \theta)$.
9: Compute $\alpha_t^{(i)} = \dfrac{f(y_t \mid X_t^{(i)}, \theta) p(X_t^{(i)} \mid X_{t-1}^{(i)}, \theta)}{q_t(X_t^{(i)} \mid X_{0:t-1}^{(i)}, y_{1:t}, \theta)}$.
10: Update $w_t^{(i)} = w_{t-1}^{(i)} \alpha_t^{(i)}$.
11: **end for**
12: Normalize weights to obtain $\tilde{w}_t^{(i)}$.
13: **end for**

In summary, the sequential updates of the importance weights together with the propagation of particles allow us to approximate the smoothing, filtering, and predictive distributions, and their expectations, in a sequential manner. The procedure is summarized in Algorithm 3.4.1. Since all of $\hat{I}_N^{\text{smth}}$, $\hat{I}_N^{\text{filt}}$ and $\hat{I}_N^{\text{pred}}$ are self-normalized, they are biased but consistent estimators. Only the marginal likelihood estimator $\hat{p}(y_{1:T} \mid \theta)$ is unbiased.

The main limitation of SIS is *weight degeneracy*. As the number of observations increases, the importance weights become increasingly unequal, because only a small fraction of the particle trajectories are able to explain all of the observations. Eventually, almost all of the normalized weight is concentrated on a few particles, and the remaining particles contribute virtually nothing to the estimates. This motivates the idea of resampling.

3.5 Sequential importance sampling with resampling

Sequential importance sampling with resampling (SISR), also known as particle filtering, is an extension of SIS that incorporates a resampling step to mitigate the problem of weight degeneracy.

Definition 3.5.1. The *effective sample size* at time t is calculated as

$$\mathrm{ESS}_t = \frac{1}{\sum_{i=1}^{N} \left(\tilde{w}_t^{(i)}\right)^2}, \tag{3.5.1}$$

where $\tilde{w}_t^{(i)}$ are the current normalized importance weights.

Resampling can be performed adaptively, only when the effective sample size falls below a chosen threshold. Note that $1 \leq \mathrm{ESS}_t \leq N$, with $\mathrm{ESS}_t = N$ for equal weights and $\mathrm{ESS}_t = 1$ when all the weight is concentrated on a single particle. In this thesis, however, resampling is performed at every time step.

Mathematically, resampling replaces the empirical distribution

$$\hat{p}_N(x_{0:t} \mid y_{1:t}) = \sum_{i=1}^{N} \tilde{w}_t^{(i)} \delta(x_{0:t} - X_{0:t}^{(i)}) \tag{3.5.2}$$

with a new empirical distribution

$$\hat{p}_N^{\mathrm{res}}(x_{0:t} \mid y_{1:t}) = \frac{1}{N} \sum_{i=1}^{N} N^{(i)} \delta(x_{0:t} - X_{0:t}^{(i)}), \tag{3.5.3}$$

where $N^{(i)}$ is the number of offspring of particle $X_{0:t}^{(i)}$ after resampling, such that $\sum_{i=1}^{N} N^{(i)} = N$. Common resampling algorithms include multinomial, stratified, residual, and systematic resampling. The last of these, systematic resampling, is used in this thesis.

The validity of resampling can be established by considering its effect on estimates of expectations with respect to the weighted empirical distribution.

As always let g be a measurable function with finite expectation. Before resampling, its expectation under the weighted empirical distribution is

$$\hat{I}_N = \sum_{i=1}^{N} \tilde{w}_t^{(i)} g(X_{0:t}^{(i)}). \tag{3.5.4}$$

After resampling, the corresponding estimate becomes

$$\hat{I}_N^{\text{res}} = \frac{1}{N} \sum_{i=1}^{N} N^{(i)} g(X_{0:t}^{(i)}). \tag{3.5.5}$$

Although $\hat{I}_N$ and $\hat{I}_N^{\text{res}}$ generally differ for a particular resampling outcome, an unbiased resampling scheme satisfies

$$E\left[N^{(i)} \mid \tilde{w}_t^{(1:N)}\right] = N\tilde{w}_t^{(i)}, \qquad i = 1, \ldots, N, \tag{3.5.6}$$

where the expectation is taken with respect to the randomness introduced by resampling. All four schemes mentioned above satisfy this condition.

Proposition 3.5.1. *Let $\hat{I}_N$ and $\hat{I}_N^{\text{res}}$ be the estimators before and after resampling. For an unbiased resampling scheme satisfying* (3.5.6)*, the resampled estimator is conditionally unbiased,*

$$E\left[\hat{I}_N^{\text{res}} \mid \tilde{w}_t^{(1:N)}\right] = \hat{I}_N, \tag{3.5.7}$$

where the expectation is taken with respect to the randomness introduced by resampling.

Proof. Since the resampling process does not depend on the particles, the values $g(X_{0:t}^{(i)})$ can be taken out of the conditional expectation, giving

$$E\left[\hat{I}_N^{\text{res}} \mid \tilde{w}_t^{(1:N)}\right] = E\left[\frac{1}{N} \sum_{i=1}^{N} N^{(i)} g(X_{0:t}^{(i)}) \,\middle|\, \tilde{w}_t^{(1:N)}\right] \tag{3.5.8}$$

$$= \frac{1}{N} \sum_{i=1}^{N} g(X_{0:t}^{(i)}) E\left[N^{(i)} \mid \tilde{w}_t^{(1:N)}\right] \tag{3.5.9}$$

$$= \frac{1}{N} \sum_{i=1}^{N} g(X_{0:t}^{(i)}) N\tilde{w}_t^{(i)} = \sum_{i=1}^{N} g(X_{0:t}^{(i)}) \tilde{w}_t^{(i)} = \hat{I}_N. \tag{3.5.10}$$

□

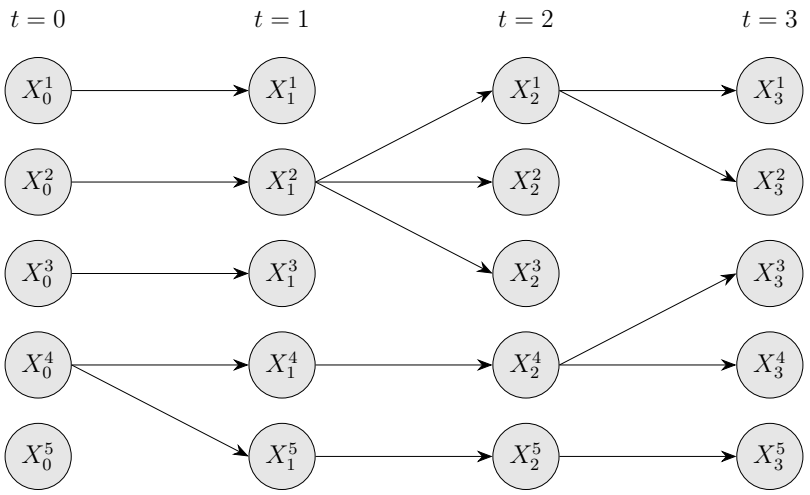


Figure 3.5.1: Schematic illustration of the resampling process. Particles with low weights are pruned, while particles with high weights are replicated.

Resampling creates a new set of equally weighted particles. A slight complication arises when keeping track of the entire trajectories of the latent states, as resampling mixes the trajectories of the particles. This can be solved by keeping track of the ancestry of the particles, which is done by storing the indices of the parent particles during resampling. This allows us to reconstruct the entire trajectory of each particle at any time step. Figure 3.5.1 illustrates the effect of resampling on the particle trajectories. The resampling step prunes particles with low weights and replicates particles with high weights, so that computational effort is concentrated on regions of high posterior probability. A side effect of repeated resampling is *path degeneracy*: going backwards in time, the reconstructed trajectories coalesce to only a few common ancestors, so the approximation of the smoothing distribution $p(x_{0:t} \mid y_{1:t})$ deteriorates for early states, even though the approximation of the filtering distribution remains accurate.

3.5.1 Bootstrap particle filter

One of the simplest and most widely used particle filtering algorithms is the bootstrap particle filter, which specifies the proposal distribution as

$$q_t(x_t \mid x_{0:t-1}, y_{1:t}, \theta) = p(x_t \mid x_{t-1}, \theta), \qquad q_0(x_0 \mid \theta) = \chi(x_0 \mid \theta), \tag{3.5.11}$$

where $p(x_t \mid x_{t-1}, \theta)$ and $\chi(x_0 \mid \theta)$ are the transition and initial densities of the state-space model. The weight update then simplifies to

$$w_t = w_{t-1} \frac{f(y_t \mid x_t, \theta) p(x_t \mid x_{t-1}, \theta)}{q_t(x_t \mid x_{0:t-1}, y_{1:t}, \theta)} = w_{t-1} f(y_t \mid x_t, \theta). \tag{3.5.12}$$

The entire bootstrap particle filter algorithm is presented in Algorithm 3.5.1. Due to the resampling step, the particles at the end of each iteration become equally weighted. Expectations with respect to the empirical distribution can be calculated either by using the weighted particles before resampling or by using the equally weighted particles after resampling. The latter introduces additional Monte Carlo variability due to the randomness of the resampling step, so the former is generally preferred. The time complexity of the bootstrap particle filter is $O(N)$ per time step, since the weight update and resampling steps each take $O(N)$ time. In total, the time complexity is $O(NT)$. Similarly, the space complexity is $O(N)$ per time step, since for every time step, we need to store N particles and their weights. But if we want entire trajectories, the space complexity is $O(NT)$, since we need to storeparticles for each time step.

Algorithm 3.5.1 Bootstrap particle filter

Require: Observations $y_{1:T}$, number of particles N, model densities χ, p, f.

for $i = 1, \ldots, N$ **do**
  Sample $X_0^{(i)} \sim \chi(x_0 \mid \theta)$.
  Set $w_0^{(i)} = \tilde{w}_0^{(i)} = 1/N$.
end for
for $t = 1, \ldots, T$ **do**
  for $i = 1, \ldots, N$ **do**
    Propagate $X_t^{(i)} \sim p(\cdot \mid X_{t-1}^{(i)}, \theta)$.
    Compute $\alpha_t^{(i)} = f(y_t \mid X_t^{(i)}, \theta)$.
    Update $w_t^{(i)} = w_{t-1}^{(i)} \alpha_t^{(i)}$.
  end for
  for $i = 1, \ldots, N$ **do**
    Normalize weights $\tilde{w}_t^{(i)} = \dfrac{w_t^{(i)}}{\sum_{j=1}^{N} w_t^{(j)}}$.
  end for
  Resample particles $\{X_t^{(i)}\}_{i=1}^N$ according to $\{\tilde{w}_t^{(i)}\}_{i=1}^N$.
  Set $w_t^{(i)} = 1/N$ for all $i = 1, \ldots, N$.
end for

3.6 Particle Markov chain Monte Carlo

So far, the model parameters θ have been assumed known. In practice they are not, and in the Bayesian framework they are treated as random variables with a prior density $p(\theta)$. The object of interest is then the posterior distribution of the parameters,

$$p(\theta \mid y_{1:T}) = \frac{p(y_{1:T} \mid \theta)\, p(\theta)}{p(y_{1:T})} \propto p(y_{1:T} \mid \theta)\, p(\theta), \tag{3.6.1}$$

where $p(y_{1:T} \mid \theta)$ is the marginal likelihood from Definition 3.4.1 and $p(y_{1:T}) = \int p(y_{1:T} \mid \theta)\, p(\theta)\, \mathrm{d}\theta$ is the (intractable) model evidence. The standard tool in such problems is Markov chain Monte Carlo (MCMC), which constructs a

Markov chain whose long-run behaviour is governed by the target distribution. For state-space models, however, MCMC faces an additional obstacle: the marginal likelihood $p(y_{1:T} \mid \theta)$ is itself intractable. Particle Markov chain Monte Carlo (PMCMC) methods [15] resolve this by combining MCMC in the parameter space with particle filters in the state space.

First, let's review the necessary Markov chain theory and the Metropolis-Hastings algorithm.

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TODO: Continue from here on. The content from here on is not yet complete. Think about if entire theory would need to be rewritten in more measure fundamental way.

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3.6.1 Markov chains

Throughout this section, let $\pi(\theta)$ denote a target density on the parameter space Θ. As in Section 3.3, the target is typically known only up to a normalizing constant, $\pi(\theta) = \tilde{\pi}(\theta)/Z$, where $\tilde{\pi}$ can be evaluated pointwise.

Definition 3.6.1. Let $(\Theta, \mathcal{B}(\Theta))$ be a measurable space. A *transition kernel* on Θ is a map $K \colon \mathcal{B}(\Theta) \times \Theta \to [0, 1]$, such that

1. for every $\theta \in \Theta$, the map $A \mapsto K(A \mid \theta)$ is a probability measure on $(\Theta, \mathcal{B}(\Theta))$;

2. for every $A \in \mathcal{B}(\Theta)$, the map $\theta \mapsto K(A \mid \theta)$ is measurable.

In other words, a transition kernel is a probability kernel from Θ to itself. The first condition says that $K(\cdot \mid \theta)$ is a probability distribution over the next state given the current state θ. This distribution may contain point masses, as is the case for the Metropolis-Hastings kernel below. The second condition ensures that quantities such as $\int K(A \mid \theta)\, \pi(\theta)\, \mathrm{d}\theta$ and $P(\theta_m \in A)$ are well defined. With a slight abuse of notation, we write $K(\theta' \mid \theta)$ for the density of $K(\cdot \mid \theta)$. A Markov chain is the sequence obtained by applying the kernel repeatedly.

Definition 3.6.2. Let K be a transition kernel on Θ. A sequence of random variables $\theta_0, \theta_1, \theta_2, \ldots$ taking values in Θ is a *Markov chain* with transition kernel K if, for every $m \geq 1$,

$$\theta_m \mid \theta_{0:m-1} \sim K(\cdot \mid \theta_{m-1}). \tag{3.6.2}$$

In words, the distribution of θ_m depends only on the previous state θ_{m-1}.

Definition 3.6.3. A distribution π is *invariant* (or stationary) for the transition kernel K if

$$\int \pi(\theta)\, K(\theta' \mid \theta)\, \mathrm{d}\theta = \pi(\theta') \quad \text{for all } \theta' \in \Theta. \tag{3.6.3}$$

Hence, if $\theta_{m-1} \sim \pi$, then also $\theta_m \sim \pi$. Verifying (3.6.3) directly requires evaluating an integral, which is rarely possible. A stronger, but local and therefore easily verifiable, condition is the following.

Definition 3.6.4. A transition kernel K satisfies *detailed balance* with respect to π if

$$\pi(\theta)\, K(\theta' \mid \theta) = \pi(\theta')\, K(\theta \mid \theta') \quad \text{for all } \theta, \theta' \in \Theta. \tag{3.6.4}$$

A Markov chain whose kernel satisfies detailed balance with respect to π is called *reversible* with respect to π.

Proposition 3.6.1. *If K satisfies detailed balance with respect to π, then π is invariant for K.*

Proof. Integrating both sides of (3.6.4) with respect to θ gives

$$\int \pi(\theta)\, K(\theta' \mid \theta)\, \mathrm{d}\theta = \pi(\theta') \int K(\theta \mid \theta')\, \mathrm{d}\theta = \pi(\theta'), \tag{3.6.5}$$

since $K(\cdot \mid \theta')$ is a probability distribution. □

Invariance alone does not guarantee that the chain actually converges to π from an arbitrary starting point. Two further properties are needed.

Definition 3.6.5. A Markov chain with transition kernel K is *π-irreducible* if for every $\theta \in \Theta$ and every measurable set A with $\pi(A) > 0$ there exists an $n \geq 1$ such that

$$P(\theta_n \in A \mid \theta_0 = \theta) > 0. \tag{3.6.6}$$

In words, from any starting point, every region that carries positive target probability is reached with positive probability after finitely many steps. Without this property the chain could remain trapped in a part of Θ and never explore the rest of the support of π.

Definition 3.6.6. A π-irreducible Markov chain with transition kernel K is said to be *periodic* with period $d \geq 2$ if there exist disjoint measurable sets $E_0, \ldots, E_{d-1} \subseteq \Theta$, each with $\pi(E_i) > 0$, such that

$$K(E_{(i+1) \bmod d} \mid \theta) = 1 \quad \text{for all } \theta \in E_i,\ i = 0, \ldots, d-1. \tag{3.6.7}$$

The chain is *aperiodic* if it is not periodic for any $d \geq 2$.

A periodic chain cycles through the sets $E_0, E_1, \ldots, E_{d-1}, E_0, \ldots$, so the distribution of θ_m keeps oscillating and cannot converge, even though time averages may still converge.

Theorem 3.6.1 (Ergodic theorem)**.** *Let $\theta_0, \theta_1, \ldots$ be a π-irreducible Markov chain with invariant distribution π. Then π is the unique invariant distribution, and the following hold for π-almost every starting point θ_0.*

1. Law of large numbers. *For every function g with $E_\pi[|g(\theta)|] < \infty$,*

$$\hat{I}_M^{\mathrm{MCMC}} = \frac{1}{M} \sum_{m=1}^{M} g(\theta_m) \xrightarrow[M \to \infty]{a.s.} E_\pi[g(\theta)]. \tag{3.6.8}$$

2. Convergence in distribution. *If the chain is in addition aperiodic, the distribution of the m-th state converges to π,*

$$\sup_A |P(\theta_m \in A \mid \theta_0) - \pi(A)| \xrightarrow[m\to\infty]{} 0. \tag{3.6.9}$$

The two statements concern different objects. The first is about the average over the whole trajectory $\theta_1, \ldots, \theta_M$ as the number of iterations M grows, and it does not require aperiodicity, since averaging over a full cycle removes any periodic behavior. The second is about the marginal distribution of a single state θ_m as its index m grows, and it justifies treating late states of the chain as approximate draws from π.

The MCMC estimator $\hat{I}_M^{\text{MCMC}}$ plays the same role as the Monte Carlo estimator (3.1.2), with the important difference that the samples are now dependent and only asymptotically distributed according to π. In practice, the first iterations (the *burn-in*) are therefore discarded.

3.6.2 Metropolis-Hastings algorithm

The Metropolis-Hastings (MH) algorithm [16, 17] is a general recipe for constructing a transition kernel that satisfies detailed balance with respect to a given target π. It requires a *proposal density* $q(\theta' \mid \theta)$, from which it is easy to sample, and which is not to be confused with the importance sampling proposals of the previous sections.

Definition 3.6.7. The *Metropolis-Hastings acceptance probability* of moving from θ to θ' is

$$\alpha(\theta, \theta') = \min\left\{1,\ \frac{\pi(\theta')\, q(\theta \mid \theta')}{\pi(\theta)\, q(\theta' \mid \theta)}\right\} = \min\left\{1,\ \frac{\tilde{\pi}(\theta')\, q(\theta \mid \theta')}{\tilde{\pi}(\theta)\, q(\theta' \mid \theta)}\right\}. \tag{3.6.10}$$

The normalizing constant Z cancels in the ratio, so, exactly as for self-normalized importance sampling, only the unnormalized density $\tilde{\pi}$ is required. The resulting procedure is given in Algorithm 3.6.1.

Algorithm 3.6.1 Metropolis-Hastings

Require: Unnormalized target $\tilde{\pi}$, proposal density q, number of iterations M, initial state θ_0 with $\tilde{\pi}(\theta_0) > 0$.

for $m = 1, \dots, M$ **do**
  Sample $\theta' \sim q(\cdot \mid \theta_{m-1})$.
  Compute $\alpha(\theta_{m-1}, \theta')$
  Sample $V \sim \mathcal{U}(0, 1)$.
  if $V \leq \alpha(\theta_{m-1}, \theta')$ **then**
    Accept: set $\theta_m = \theta'$.
  else
    Reject: set $\theta_m = \theta_{m-1}$.
  end if
end for

A proposed move is accepted with probability $\alpha(\theta, \theta')$; otherwise the chain stays where it is. The probability of staying at θ is therefore

$$r(\theta) = 1 - \int \alpha(\theta, \theta')\, q(\theta' \mid \theta)\, \mathrm{d}\theta', \tag{3.6.11}$$

and the transition kernel of Algorithm 3.6.1 is

$$K_{\mathrm{MH}}(\theta' \mid \theta) = \alpha(\theta, \theta')\, q(\theta' \mid \theta) + r(\theta)\, \delta(\theta' - \theta). \tag{3.6.12}$$

The first term corresponds to accepted proposals, and the second term, a point mass at the current state, to rejections.

Theorem 3.6.2. *The Metropolis-Hastings kernel* (3.6.12) *satisfies detailed balance with respect to* π*. Consequently,* π *is an invariant distribution of the Metropolis-Hastings chain.*

Proof. Multiplying (3.6.12) by $\pi(\theta)$ gives

$$\pi(\theta)\, K_{\mathrm{MH}}(\theta' \mid \theta) = \pi(\theta)\, \alpha(\theta, \theta')\, q(\theta' \mid \theta) + \pi(\theta)\, r(\theta)\, \delta(\theta' - \theta). \tag{3.6.13}$$

We show that each of the two terms on the right-hand side is unchanged when θ and θ' are interchanged.

For the first term, we have

$$\pi(\theta)\, q(\theta' \mid \theta)\, \min\left\{1,\, \frac{\pi(\theta')\, q(\theta \mid \theta')}{\pi(\theta)\, q(\theta' \mid \theta)}\right\} = \min\left\{\pi(\theta)\, q(\theta' \mid \theta),\, \pi(\theta')\, q(\theta \mid \theta')\right\}. \tag{3.6.14}$$

The right-hand side is symmetric in θ and θ', hence

$$\pi(\theta)\, \alpha(\theta, \theta')\, q(\theta' \mid \theta) = \pi(\theta')\, \alpha(\theta', \theta)\, q(\theta \mid \theta'). \tag{3.6.15}$$

Adding both terms gives $\pi(\theta)\, K_{\mathrm{MH}}(\theta' \mid \theta) = \pi(\theta')\, K_{\mathrm{MH}}(\theta \mid \theta')$. Invariance follows from Proposition 3.6.1. □

The theorem holds for any proposal density q. However, invariance alone is not enough for Theorem 3.6.1, which also requires irreducibility and aperiodicity, and these do depend on q. A simple sufficient condition is that $q(\theta' \mid \theta) > 0$ for all θ, θ' in the support of π. Then for every θ with $\pi(\theta) > 0$ and every set A with $\pi(A) > 0$, the acceptance probability $\alpha(\theta, \theta')$ is positive for all $\theta' \in A$ with $\pi(\theta') > 0$, so

$$K_{\mathrm{MH}}(A \mid \theta) \geq \int_A \alpha(\theta, \theta')\, q(\theta' \mid \theta)\, \mathrm{d}\theta' > 0. \tag{3.6.16}$$

Every set of positive target probability is therefore reached in a single step, which gives π-irreducibility with $n = 1$ in Definition 3.6.5. It also rules out periodicity. If sets $E_0, \ldots, E_{d-1}$ as in Definition 3.6.6 existed, then for $\theta \in E_0$ we would have $K_{\mathrm{MH}}(E_0 \mid \theta) > 0$, contradicting $K_{\mathrm{MH}}(E_1 \mid \theta) = 1$ since E_0 and E_1 are disjoint. Beyond these qualitative properties, the choice of q determines how quickly the chain explores Θ and hence how many iterations are needed in practice. A common choice is the Gaussian random walk proposal $q(\theta' \mid \theta) = \mathcal{N}(\theta'; \theta, \Sigma)$, which is symmetric, $q(\theta' \mid \theta) = q(\theta \mid \theta')$, so that the proposal densities cancel in (**??**).

Applied to Bayesian parameter inference with target (3.6.1), we have $\tilde{\pi}(\theta) = p(y_{1:T} \mid \theta)\, p(\theta)$ and $Z = p(y_{1:T})$, so the acceptance probability becomes

$$\alpha(\theta, \theta') = \min\left\{1,\, \frac{p(y_{1:T} \mid \theta')\, p(\theta')\, q(\theta \mid \theta')}{p(y_{1:T} \mid \theta)\, p(\theta)\, q(\theta' \mid \theta)}\right\}. \tag{3.6.17}$$

The intractable evidence $p(y_{1:T})$ cancels, but the marginal likelihood $p(y_{1:T} \mid \theta)$ does not. For the MSSV model, and for nonlinear state-space models in general, Algorithm 3.6.1 therefore cannot be implemented directly. A natural idea is to replace $p(y_{1:T} \mid \theta)$ in (3.6.17) by a particle filter estimate. Remarkably, if this is done correctly, the resulting chain still targets the exact posterior. The key property that makes this possible is unbiasedness of the estimator, established in Theorem **??**.

3.6.3 Pseudo-marginal Metropolis-Hastings

We now formalize what it means to replace the marginal likelihood in (3.6.17) by an estimate. Consider the following general setting [18]. For each θ, let U denote a collection of auxiliary random variables with distribution $\psi(u \mid \theta)$, and let $\hat{p}(y_{1:T} \mid \theta, u)$ be an estimator of the marginal likelihood computed from $U = u$, satisfying

$$\hat{p}(y_{1:T} \mid \theta, u) \geq 0 \qquad \text{and} \qquad \int \hat{p}(y_{1:T} \mid \theta, u)\, \psi(u \mid \theta)\, \mathrm{d}u = p(y_{1:T} \mid \theta). \tag{3.6.18}$$

The *pseudo-marginal Metropolis-Hastings* algorithm, given in Algorithm 3.6.2, is Algorithm 3.6.1 with $p(y_{1:T} \mid \theta)$ replaced by its estimate. It is crucial that the estimate associated with the current state is stored and reused in subsequent iterations, rather than recomputed. Recomputing it would yield a different algorithm that does not, in general, target the correct distribution.

Algorithm 3.6.2 Pseudo-marginal Metropolis-Hastings

Require: Prior $p(\theta)$, proposal density q, auxiliary distribution ψ, estimator $\hat{p}$, number of iterations M, initial state θ_0.

1: Sample $U_0 \sim \psi(\cdot \mid \theta_0)$ and compute $\hat{p}_0 = \hat{p}(y_{1:T} \mid \theta_0, U_0)$; repeat until $p(\theta_0)\,\hat{p}_0 > 0$.
2: **for** $m = 1, \ldots, M$ **do**
3: Sample $\theta' \sim q(\cdot \mid \theta_{m-1})$.
4: Sample $U' \sim \psi(\cdot \mid \theta')$ and compute $\hat{p}' = \hat{p}(y_{1:T} \mid \theta', U')$.
5: Compute

$$\hat{\alpha} = \min\left\{1,\ \frac{\hat{p}'\,p(\theta')\,q(\theta_{m-1} \mid \theta')}{\hat{p}_{m-1}\,p(\theta_{m-1})\,q(\theta' \mid \theta_{m-1})}\right\}.$$

6: Sample $V \sim \mathcal{U}(0,1)$.
7: **if** $V \leq \hat{\alpha}$ **then**
8: Accept: set $(\theta_m, \hat{p}_m) = (\theta', \hat{p}')$.
9: **else**
10: Reject: set $(\theta_m, \hat{p}_m) = (\theta_{m-1}, \hat{p}_{m-1})$.
11: **end if**
12: **end for**

At first sight, Algorithm 3.6.2 appears to be merely an approximation of Algorithm 3.6.1. The key insight is that it is an *exact* Metropolis-Hastings algorithm, but on an extended space that includes the auxiliary variables.

Definition 3.6.8. The *extended target* on the space of (θ, u) is defined as

$$\bar{\pi}(\theta, u) = \frac{p(\theta)\,\hat{p}(y_{1:T} \mid \theta, u)\,\psi(u \mid \theta)}{p(y_{1:T})}. \tag{3.6.19}$$

Theorem 3.6.3. *Assume that the estimator satisfies* (3.6.18)*. Then:*

1. *$\bar{\pi}$ is a probability density on the extended space;*
2. *the marginal of $\bar{\pi}$ in θ is the posterior distribution, $\int \bar{\pi}(\theta, u)\,\mathrm{d}u = p(\theta \mid y_{1:T})$;*

3. the sequence $(\theta_m, U_m)_{m\geq 0}$ generated by Algorithm 3.6.2, where U_m denotes the auxiliary variables associated with the accepted estimate $\hat{p}_m$, is a Metropolis-Hastings chain with target $\bar{\pi}$ and proposal

$$Q(\theta', u' \mid \theta, u) = q(\theta' \mid \theta)\, \psi(u' \mid \theta'). \tag{3.6.20}$$

Consequently, $\bar{\pi}$ is invariant for $(\theta_m, U_m)_{m\geq 0}$, and if $(\theta_{m-1}, U_{m-1}) \sim \bar{\pi}$, then $\theta_m \sim p(\theta \mid y_{1:T})$.

Proof. 1. Clearly $\bar{\pi} \geq 0$. Integrating first over u and using (3.6.18),

$$\iint \bar{\pi}(\theta, u)\, \mathrm{d}u\, \mathrm{d}\theta = \frac{1}{p(y_{1:T})} \int p(\theta) \left(\int \hat{p}(y_{1:T} \mid \theta, u)\, \psi(u \mid \theta)\, \mathrm{d}u \right) \mathrm{d}\theta = \frac{\int p(\theta)\, p(y_{1:T} \mid \theta)\, \mathrm{d}\theta}{p(y_{1:T})} = 1. \tag{3.6.21}$$

2. Again by (3.6.18) and by (3.6.1),

$$\int \bar{\pi}(\theta, u)\, \mathrm{d}u = \frac{p(\theta)\, p(y_{1:T} \mid \theta)}{p(y_{1:T})} = p(\theta \mid y_{1:T}). \tag{3.6.22}$$

3. Consider Algorithm 3.6.1 on the extended space with target $\bar{\pi}$ and proposal Q. Proposing from Q amounts to sampling $\theta' \sim q(\cdot \mid \theta)$ and then $u' \sim \psi(\cdot \mid \theta')$, which is precisely what Algorithm 3.6.2 does. The Metropolis-Hastings ratio on the extended space is

$$\frac{\bar{\pi}(\theta', u')\, Q(\theta, u \mid \theta', u')}{\bar{\pi}(\theta, u)\, Q(\theta', u' \mid \theta, u)} = \frac{p(\theta')\, \hat{p}(y_{1:T} \mid \theta', u')\, \psi(u' \mid \theta')\; q(\theta \mid \theta')\, \psi(u \mid \theta)}{p(\theta)\, \hat{p}(y_{1:T} \mid \theta, u)\, \psi(u \mid \theta)\; q(\theta' \mid \theta)\, \psi(u' \mid \theta')} \tag{3.6.23}$$

$$= \frac{\hat{p}(y_{1:T} \mid \theta', u')\, p(\theta')\, q(\theta \mid \theta')}{\hat{p}(y_{1:T} \mid \theta, u)\, p(\theta)\, q(\theta' \mid \theta)}, \tag{3.6.24}$$

since the evidence $p(y_{1:T})$ and the densities of the auxiliary variables ψ cancel. This is exactly the ratio in $\hat{\alpha}$ of Algorithm 3.6.2, where $\hat{p}_{m-1} = \hat{p}(y_{1:T} \mid \theta_{m-1}, U_{m-1})$. Moreover, on acceptance both θ and u are replaced, and on rejection both are kept. Since the algorithm uses u only through the value $\hat{p}(y_{1:T} \mid \theta, u)$, storing $\hat{p}_m$ is equivalent to storing U_m. Hence Algorithm 3.6.2 is Algorithm 3.6.1 on the extended space.

By Theorem 3.6.2, the chain $(\theta_m, U_m)_{m\geq 0}$ satisfies detailed balance with respect to $\bar{\pi}$, so $\bar{\pi}$ is invariant. If $(\theta_{m-1}, U_{m-1}) \sim \bar{\pi}$, then $(\theta_m, U_m) \sim \bar{\pi}$, and by the second claim the marginal distribution of θ_m is $p(\theta \mid y_{1:T})$. □

The proof also explains why the estimate of the current state must not be recomputed. Reusing $\hat{p}_{m-1}$ corresponds to keeping U_{m-1} as part of the state of the chain, whereas drawing a fresh U for the current state at every iteration would alter the transition kernel, and the resulting kernel no longer satisfies detailed balance with respect to $\bar{\pi}$. The proof also shows that unbiasedness is exactly the property needed. Without it, the θ-marginal of $\bar{\pi}$ would be proportional to $p(\theta)\, E_\psi[\hat{p}(y_{1:T} \mid \theta, U)]$ instead of to the posterior.

3.6.4 Particle marginal Metropolis-Hastings

Particle marginal Metropolis-Hastings (PMMH) [15] is the pseudo-marginal Metropolis-Hastings algorithm in which the estimator is the particle filter marginal likelihood estimator (**??**). Here the auxiliary variables U consist of all the random variables generated by one run of the bootstrap particle filter at parameter θ, that is, the initial particles, the propagated particles at every time step and the outcomes of all resampling steps. Their joint distribution $\psi(u \mid \theta)$ is implicitly defined by Algorithm 3.5.1 and never needs to be evaluated, since it cancels in the acceptance ratio. The complete procedure is given in Algorithm 3.6.3.

Algorithm 3.6.3 Particle marginal Metropolis-Hastings

Require: Observations $y_{1:T}$, number of particles N, prior $p(\theta)$, proposal density q, number of iterations M, initial state θ_0.

1: Run Algorithm 3.5.1 at θ_0 and compute $\hat{p}_0 = \hat{p}^{\mathrm{PF}}(y_{1:T} \mid \theta_0)$.
2: **for** $m = 1, \ldots, M$ **do**
3: Sample $\theta' \sim q(\cdot \mid \theta_{m-1})$.
4: Run Algorithm 3.5.1 at θ' and compute $\hat{p}' = \hat{p}^{\mathrm{PF}}(y_{1:T} \mid \theta')$.
5: Compute

$$\hat{\alpha} = \min\left\{1, \frac{\hat{p}'\, p(\theta')\, q(\theta_{m-1} \mid \theta')}{\hat{p}_{m-1}\, p(\theta_{m-1})\, q(\theta' \mid \theta_{m-1})}\right\}.$$

6: Sample $V \sim \mathcal{U}(0,1)$.
7: **if** $V \leq \hat{\alpha}$ **then**
8: Accept: set $(\theta_m, \hat{p}_m) = (\theta', \hat{p}')$.
9: **else**
10: Reject: set $(\theta_m, \hat{p}_m) = (\theta_{m-1}, \hat{p}_{m-1})$.
11: **end if**
12: **end for**

Theorem 3.6.4. *For every number of particles $N \geq 1$, the particle marginal Metropolis-Hastings chain admits an invariant distribution whose marginal in θ is the exact posterior distribution $p(\theta \mid y_{1:T})$.*

Proof. The estimator $\hat{p}^{\mathrm{PF}}(y_{1:T} \mid \theta)$ is a product of averages of densities and is therefore non-negative. By Theorem **??**, it is unbiased for every $N \geq 1$ and every θ, that is,

$$\int \hat{p}^{\mathrm{PF}}(y_{1:T} \mid \theta, u)\, \psi(u \mid \theta)\, \mathrm{d}u = p(y_{1:T} \mid \theta), \tag{3.6.25}$$

where the dependence on the particle filter randomness u has been made explicit. Condition (3.6.18) is thus satisfied, and Algorithm 3.6.3 is a special case of Algorithm 3.6.2. The claim follows from Theorem 3.6.3. □

Remark 3.6.1. The auxiliary variables of the particle filter include discrete resampling indices, so $\psi(u \mid \theta)$ is in general not a density with respect to

the Lebesgue measure. As with the Dirac delta notation used throughout this chapter, the integrals over u should be understood with respect to the appropriate probability measure. The proofs of Theorems 3.6.2 and 3.6.3 carry over without change, since ψ only ever appears as the distribution from which u' is proposed and cancels in the acceptance ratio.

Theorem 3.6.4 shows that PMMH is not an approximation: the number of particles does not affect the invariant distribution, only how efficiently the chain explores it. Under mild additional conditions, essentially that the corresponding exact Metropolis-Hastings chain would be irreducible and aperiodic, the PMMH chain also converges to its invariant distribution [15]. The role of N is best understood by noting that the acceptance ratio of PMMH equals the ideal ratio (3.6.17) multiplied by

$$\frac{\hat{p}^{\mathrm{PF}}(y_{1:T} \mid \theta')/p(y_{1:T} \mid \theta')}{\hat{p}^{\mathrm{PF}}(y_{1:T} \mid \theta)/p(y_{1:T} \mid \theta)}. \tag{3.6.26}$$

If the estimate at the current state happens to be much larger than the true likelihood, this factor is small for most proposals, and the chain may get stuck at the same value for many iterations. A small N produces noisy estimates and hence poor mixing, while a large N makes every iteration expensive. As a rule of thumb, [19] recommend that N should be chosen such that the standard deviation of $\log \hat{p}^{\mathrm{PF}}(y_{1:T} \mid \theta)$ around the posterior mode is approximately between 1 and 1.7.

In practice, the acceptance ratio is computed on the logarithmic scale to avoid numerical underflow, since $\hat{p}^{\mathrm{PF}}(y_{1:T} \mid \theta)$ is a product of T factors. Moreover, if the proposed θ' lies outside the support of the prior, then $p(\theta') = 0$ and the proposal can be rejected immediately, without running the particle filter.

Remark 3.6.2. Algorithm 3.6.3 targets only the posterior distribution of the parameters. It is shown in [15] that PMMH can be extended to target the joint posterior $p(\theta, x_{0:T} \mid y_{1:T})$ as well. Each time the particle filter is run at a proposed θ', a trajectory $X'_{0:T}$ is additionally drawn from its empirical

smoothing distribution, and this trajectory is accepted or rejected together with θ'. The acceptance probability is unchanged.

3.6.5 Particle Gibbs

3.7 SMC^2

Chapter 4

Implementation and Evaluation Methodology

The source code accompanying this thesis is publicly available on GitHub: `https://github.com/vitolev/mssv-smc`. The repository contains the implementation of the MSSV model and inference algorithms, as well as experiment scripts, configuration files, tests, and notebooks used to generate the results presented in this thesis.

4.1 Implementation of the MSSV model and inference algorithms

The implementation follows a layered design. The common interfaces are defined in `src/models/base.py`. `StateSpaceModel` specifies the operations needed by all inference algorithms: sampling the initial and next states, and evaluating observation and transition densities. The associated abstract class `StateSpaceModelParams` defines the container of parameters of the model, class `StateSpaceModelState` defines the state structure and operations on it, class `StateSpaceModelPrior` defines the prior of the model parameters and its densities and `StateSpaceModelProposal` defines the proposal distributions for the parameters, used in the PMMH, PG and SMC^2 algorithms.

Consequently, the filters do not depend on the details of a particular state-space model.

The MSSV model is implemented by the classes in `src/models/mssv.py`. `MSSVParams` is an immutable parameter container. It reconstructs the ordered regime means from μ_1 and Δ_i, validates the transition matrix, and provides constrained and unconstrained vector representations and conversions between them. `MSSVState` stores the continuous log-volatility `h_t` and the discrete regime `s_t` together, so that a particle is always a complete state. `MSSVModel` implements the model specific sampling and density methods required by `StateSpaceModel`. Its `MSSVPrior` supplies initial parameter particles, while `MSSVProposal` supplies random walk, independent, or trajectory conditional parameter updates.

Particle filtering is represented by the abstract `ParticleFilter` class in `src/filters/smc/base_pf.py`. It owns the model, particle count, random-number generator, and resampling strategy, and also provides backward reconstruction of smoothing trajectories from the stored particle history. `BootstrapParticleFilter` implements the standard bootstrap filter (Algorithm 3.5.1) and its conditional version, which fixes one particle to a reference trajectory. Resampling itself is kept independent in `resampling.py`. Each filter history stores particles, normalized weights, ancestor indices, and the accumulated log marginal likelihood.

The PMCMC layer composes these objects rather than duplicating model logic. The PMMH is defined in `src/pmcmc/pmmh.py`. Inside `PMMH_Chain` owns one parameter chain. At initialization it samples `MSSVParams` from `MSSVPrior`, runs the particle filter, and stores one smoothing trajectory [1]. Each subsequent step asks `MSSVProposal` for a parameter proposal, runs the same particle filter again, and accepts or rejects the proposed parameter and trajectory together. Entire experiment facing wrapper is defined as `ParticleMarginalMetropolisHastings`: it creates independent random-

[1] Actually, it samples from general definition of `StateSpaceModel`, as the implementation supports any state-space model.

number streams, runs multiple `PMMH_Chain` instances in parallel, and writes each chain to an HDF5 file. Similar structure is used for PG, except that `PGS_Chain` in `src/pmcmc/pg.py` uses the conditional particle filter instead. After each conditional filter run it samples a new trajectory and passes that trajectory to the conditional proposal in `MSSVProposal`. The entire wrapper is then defined as `ParticleGibbsSampler`, with a similar functionality as in PMMH.

`SMC2` in `src/filters/smc2/smc2.py` adds a second particle population around the existing particle filter. A `ThetaParticle` contains one parameter value, its associated state particles, and its parameter weight. At every observation, `SMC2` propagates the state particles and adds their incremental particle filter likelihood to the parameter weight. When the parameter effective sample size falls below the configured threshold, parameter particles are resampled and rejuvenated using `MSSVProposal`. The parameter histories, effective sample sizes, resampling times, state particles and final smoothing trajectories are stored in the separate HDF5 files.

The scripts in `experiments/` run the experiments. They read the settings from a YAML file, load the data, create the MSSV model and bootstrap particle filter, and then start PMMH, PG, or SMC^2. The same source code is used for synthetic, Bitcoin, and S&P 500 data; only the data and settings change. Separate utilities generate synthetic data and produce diagnostic plots. The scripts also set the random seed, logging, and output directories.

4.2 Datasets

4.2.1 Synthetic Data

The synthetic data is generated from the MSSV model with known parameters. We generated six datasets, each with two regimes and $T = 2000$

observations. The first three datasets share the parameters

$$\mu_1 = -5, \qquad \mu_2 = -2$$
$$\sigma^2 = 0.1, \qquad P = \begin{pmatrix} 0.99 & 0.01 \\ 0.015 & 0.985 \end{pmatrix},$$

but have different persistence parameters: $\phi = 0.9$, $\phi = 0.7$, and $\phi = 0.5$, respectively. The remaining three datasets use

$$\mu_1 = -1, \qquad \mu_2 = 1$$
$$\sigma^2 = 0.05, \qquad P = \begin{pmatrix} 0.99 & 0.01 \\ 0.05 & 0.95 \end{pmatrix},$$

with the same three values of ϕ. This produces datasets with different regime separation, transition probabilities, and persistence, allowing the robustness of the methods to be examined under several controlled settings.

4.2.2 Bitcoin Data

The Bitcoin data was obtained from the *Bitcoin Historical Data* dataset published by Zielak on Kaggle [20]. The dataset contains Bitcoin market data at one-minute intervals, including open, high, low, close, and volume fields spanning from January 1, 2012 to July 16, 2026. For every day we calculated its close-to-close log-return, that is being used as the input for the MSSV model. The first observation, for which no preceding close exists, was removed. In total this gives us $T = 5310$ daily log-returns.

As the Bitcoin market is open 24/7 and we have access to intraday data, we can calculate its daily realized variance $RV_t^{(288)}$, where 288 is the number of 5-minute intervals in a day.

For prior elicitation we used another dataset of historical daily Bitcoin data obtained from CoinMarketCap's Bitcoin historical-data page [21]. This dataset covers the period from July 13, 2010 to December 31, 2011. It was used only for prior elicitation and was not combined with the Kaggle series.

4.2.3 S&P 500 Data

Daily S&P 500 data (ticker symbol `^SPX`) was obtained from Yahoo Finance's historical-data page [22]. The dataset contains daily open, high, low, close, and volume fields spanning from December 30, 1927 to July 2, 2026. We calculated close-to-close log-returns for every day, but because the market is closed on weekends and holidays, we end up with less observations than there is calendar days.

To obtain the realized-variance series, we used the Oxford-Man Institute realized volatility indices [23]. The data were retrieved from the archived download of the Oxford-Man Institute website. The data contains already calculated realized variances for the S&P 500 index, based on 5-minute intraday returns, from January 3, 2000 to February 25, 2022. Additionally, we managed to find a small dataset of 1 minute intraday values of S&P 500 index from September 30, 2022 to September 29, 2023, which we used to calculate the realized variance for that period. Using these two datasets we calculated factor

$$c = \frac{\frac{1}{T}\sum_{t=1}^{T}\left(Y_t - \hat{\mu}_t\right)^2}{\frac{1}{T}\sum_{t=1}^{T} RV_t^{(m)}} = 1.4254$$

to account for the overnight effect, as described in Section 2.1.2.

The model was then fitted on the daily data from January 3, 2000 onwards, as we have the the realized variance for most of that period. This gives us $T = 6663$ observations. The short period when the realized variance is not available was still used for fitting, but excluded from the evaluation of the latent volatility. The daily data before the fiting period was then used for prior elicitation.

4.3 Prior elicitation

We already defined and specified the prior distributions and their hyperparameters in Section 2.4.1. The only hyperparameter that was not specified is the mean value of first regime a_0 in the prior of μ_1. For synthetic data we

set a_0 to the average of the means of both regimes, which is -3.5.

For both the Bitcoin and S&P 500 data, we used the data before the start of the fitting period for prior elicitation. Specifically, we calculated the standard deviation σ_{prior} of the log-returns in the corresponding prior dataset and set the mean of the first regime to the logarithm of the corresponding variance,

$$a_0 = 2\log\left(\sigma_{\text{prior}}\right) = \log\left(\sigma_{\text{prior}}^2\right).$$

Thus, we set the prior mean of the first regime to the average historical volatility. For Bitcoin data this results in

$$\sigma_{\text{prior}} = 0.04098, \qquad a_0 = -6.3896,$$

while for S&P 500 data we get

$$\sigma_{\text{prior}} = 0.01216, \qquad a_0 = -8.8192.$$

4.4 Evaluation methodology

The evaluation has two complementary objectives. First, the synthetic experiments assess whether the algorithms recover known parameters and latent states. Second, the Bitcoin and S&P 500 experiments explore the smoothing, filtering, and predictive behaviour and assess whether the inferred latent volatility and regimes are consistent with realized variance and with known economic events.

4.4.1 Synthetic data evaluation

Since the data-generating parameters and latent trajectories are known, they provide reference values for evaluating parameter and state inference.

For each model parameter, posterior summaries are compared with the corresponding data-generating value. We report the posterior mean and central 95% credible interval, and assess whether the true value is contained within the interval.

Since h_t represents the log variance, with $\sigma_t^2 = \exp(h_t)$, there are several possible ways to evaluate the accuracy of the inferred volatility. We focus on variance rather than volatility or log variance because realized variance, which is used as a proxy for the latent variance in the financial data application, is measured on the variance scale. We consider the following two losses:

$$\text{MSE} = \frac{1}{T}\sum_{t=1}^{T}\left(\hat{\sigma}_t^2 - \sigma_t^2\right)^2,$$

$$\text{QLIKE} = \frac{1}{T}\sum_{t=1}^{T}\left(\log\hat{\sigma}_t^2 + \frac{\sigma_t^2}{\hat{\sigma}_t^2}\right),$$

where $\hat{\sigma}_t^2$ denotes the estimated variance, while σ_t^2 denotes the true variance. The first loss is the mean squared error of the variance, and the second is the QLIKE loss, a commonly used loss for evaluating volatility models.

For PMMH and PG, which are offline methods, we can evaluate only the smoothing distribution $p(h_{1:T}, s_{1:T} \mid y_{1:T})$, obtained after processing the complete dataset. The point estimate of the variance is obtained by transforming the posterior distribution and taking the expectations

$$\hat{\sigma}_{t|T}^2 = E[\exp(h_t) \mid y_{1:T}].$$

Using the smoothing estimates, we calculate the two losses for the entire time series.

In addition to smoothing, SMC^2 allows us to evaluate filtering and one-step-ahead predictive performance sequentially. We therefore report the two losses for the smoothing estimates over the entire time series, as well as the evolution of the filtering and predictive losses over time.

For filtering, the posterior distribution at time t is given by $p(h_t, s_t \mid y_{1:t})$. The corresponding point estimate of the variance then

$$\hat{\sigma}_{t|t}^2 = E[\exp(h_t) \mid y_{1:t}].$$

At each time step, we compare these estimates with the corresponding true values and calculate the two losses. This allows us to examine how the accu-

racy of the filtered volatility estimates evolves as new observations become available.

Similarly, SMC^2 enables us to compute the one-step-ahead predictive distributions $p(h_{t+1}, s_{t+1} \mid y_{1:t})$. The predictive point estimate is obtained as

$$\hat{\sigma}^2_{t+1|t} = E[\exp(h_{t+1}) \mid y_{1:t}].$$

These estimates are compared with the corresponding true values at time $t+1$, allowing us to evaluate the one-step-ahead predictive performance of the model as new observations become available.

Finally, for the latent regime, we evaluate the smoothing estimates. We report the posterior probabilities of each regime, as well as the posterior mode

$$\hat{s}_{t|T} = \arg \max_{k \in \{1,\dots,K\}} P(s_t = k \mid y_{1:T}).$$

Comparing the inferred regimes with the true regime sequence $s_{1:T}$, we calculate the proportion of correctly classified regimes.

4.4.2 Evaluation on financial data

Unlike the synthetic data experiments, the true parameter values and latent trajectories are unknown for real-world data. Consequently, parameter inference cannot be evaluated against known generating values, nor can the accuracy of the inferred latent volatility and regime sequences be assessed directly.

For model parameters, we therefore report posterior summaries, consisting of the posterior mean and central 95% credible interval, to describe the uncertainty associated with the parameter estimates.

For the latent volatility, we evaluate the performance using the same two losses as in the synthetic data evaluation, but now the true variance is replaced by a proxy measure of realized variance, as justified in Section 2.1.1.

Specifically,

$$\mathrm{MSE}_{RV} = \frac{1}{T}\sum_{t=1}^{T}\left(\hat{\sigma}_t^2 - RV_t\right)^2,$$

$$\mathrm{QLIKE}_{RV} = \frac{1}{T}\sum_{t=1}^{T}\left(\log\hat{\sigma}_t^2 + \frac{RV_t}{\hat{\sigma}_t^2}\right).$$

Since realized variance is an imperfect proxy for the latent variance, these losses should be interpreted as measures of agreement between the inferred variance and the realized variance proxy, rather than as direct measures of the accuracy of the latent variance estimates.

The workflow for evaluating PMMH and PG is similar to that of the synthetic data and we only report the losses for the smoothing estimates over the entire time series. For SMC^2, we additionally report the evolution of the filtering and predictive losses over time, just as in the synthetic data evaluation.

Finally, for the latent regime, we report the posterior probabilities of each regime and the posterior mode obtained from the smoothing distributions. Since the regimes are unobservable modelling assumptions, there is no direct way to evaluate the accuracy of the inferred regime sequence. However, we can examine the timing of inferred regime changes and compare them with known economic events, such as financial crises or market shocks, to assess whether the identified regimes are consistent with real-world market conditions.

4.5 Simulation settings

The computational settings were chosen before running the final experiments and were kept fixed within each data group. For PMMH and PG, the number of particles refers to the bootstrap particle filter used inside every PMCMC iteration. The synthetic data experiments used $N = 2000$ particles, whereas the both real data experiments used $N = 8000$ particles. The larger particle population reduces the variance of the particle filter likelihood estimate and

is therefore particularly important for PMMH, whose acceptance probability depends directly on this estimate. The same particle counts were used for PG to make the comparison between the two PMCMC methods more consistent.

Each synthetic data PMMH and PG experiment consisted of $C = 16$ independent parallel chains. Each chain was run for $M = 10000$ iterations, of which the first 5000 were discarded as burn-in. On the other hand, the real data experiments used $C = 32$ chains, $M = 20000$ iterations per chain, and a burn-in of 10000 iterations. Thus, the retained output contains 5000 draws per chain for synthetic data and 10000 draws per chain for each real dataset. Across all chains this gives 80000 and 320000 retained draws, respectively.

For the proposal distribution $q(\theta^* \mid \theta)$ used in PMMH step update, we use random walk on the unconstrained version of parameters such that $\theta^* = \theta + \varepsilon$, where $\varepsilon \sim \mathcal{N}(0, \Sigma)$ where Σ is a covariance matrix chosen as

$$\Sigma = \begin{bmatrix} 0.0400 & -0.0054 & -0.0029 & 0.0007 & 0.0136 & -0.0057 \\ -0.0054 & 0.0225 & 0.0046 & -0.0041 & -0.0079 & 0.0147 \\ -0.0029 & 0.0046 & 0.0100 & -0.0031 & -0.0031 & 0.0032 \\ 0.0007 & -0.0041 & -0.0031 & 0.0100 & 0.0047 & -0.0046 \\ 0.0136 & -0.0079 & -0.0031 & 0.0047 & 0.0900 & -0.0320 \\ -0.0057 & 0.0147 & 0.0032 & -0.0046 & -0.0320 & 0.0900 \end{bmatrix}$$

This matrix was obtained by first running a shorter prior experiment and noting the correlation matrix of posterior samples R. Then we manually chose the diagonal matrix of standard deviations D, such that we obtained reasonable step size for the random walk. The final covariance matrix is then $\Sigma = DRD$. The same covariance matrix was used for both synthetic and real data.

SMC^2 uses a different computational structure and consequently has no separate burn-in or user-specified number of MCMC iterations. It processes the data sequentially for all T observations. Every parameter particle contains an inner particle filter with N_x state particles. The outer population contains N_θ parameter particles. For synthetic data we used $(N_x, N_\theta) = (2000, 4000)$, while both real-data experiments used $(N_x, N_\theta) = (8000, 8000)$.

Parameter particles were resampled whenever their effective sample size fell below γN_θ, with $\gamma = 0.5$. After each such resampling event, $R = 10$ MCMC rejuvenation steps were applied to every parameter particle. The value `x_save_factor`=5 was used only to control how many state particles were stored in the HDF5 history, to reduce the size of the output files.

All experiments were executed on the SLING computing cluster on nodes with AMD EPYC 9745 (128c, 2.4 GHz) processor and 1TB of RAM.

Chapter 5

Results

5.1 Synthetic data

5.2 Bitcoin data

5.3 S&P 500 data

Appendix A

Title of the appendix 1

Example of the appendix.

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